BANKING ANALYTICS / RESEARCH PAPER

Customer Engagement & Product Utilization Analytics for Retention Strategy

An observational study of customer activity, product breadth and financial commitment

Table 1: Research at a glance

Population in supplied outputs	Observed churn	Observed retention
10,000 customer observations	20.37% (2,037)	79.63% (7,963)

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Evidence edition: supplied analytical outputs and dashboard screenshots

Evidence boundary: the raw customer CSV, Colab notebook, generated-output ZIP and Streamlit source code were not supplied. Findings are verified within the available chart data, not against an independently audited customer database. No affiliation or qualifications are inferred.

Abstract

Background: Retention analysis in retail banking requires distinguishing observable activity and product holdings from actual customer engagement. Financial balances and product count may describe a relationship without establishing its quality or durability.

Objective: This study examines associations between churn, customer activity, product breadth, financial balances, card ownership and supporting characteristics, and assesses transparent segmentation and relationship scoring.

Methods: Twelve supplied Plotly HTML outputs and six authentic dashboard screenshots were examined. Aggregate counts and rates were extracted and reconciled across outputs. A balance–activity–outcome figure contained 10,000 plotted observations, allowing supplementary financial summaries. Newly computed Wilson confidence intervals, Pearson chi-square tests, Cramér's V and selected rate contrasts were distinguished from source results. Nine omnibus tests were adjusted using Holm's procedure.

Results: The outputs contain 2,037 churned and 7,963 retained observations, corresponding to 20.37% churn. Inactive customers had 26.85% churn versus 14.27% among active customers, a 12.58-percentage-point difference and a rate ratio of 1.88. Churn was non-linear across product counts: 27.71%, 7.58%, 82.71% and 100.00% for one through four products, respectively. Inactive customers above the reconstructed balance 80th percentile numbered 985, with 30.05% churn. Cardholder retention exceeded non-cardholder retention by only 0.63 percentage points, with an interval spanning zero. Churn decreased across the four reported relationship-score bands.

Implications and limitations: Segment review and controlled evaluation are preferable to automatic cross-selling or score-based targeting. These observational outputs support descriptive associations, not intervention effects or validated churn predictions. Raw-data quality, reporting period, score implementation and complete dashboard behavior remain unverified because underlying project files were unavailable.

Keywords: customer retention; customer churn; customer engagement; product breadth; banking analytics; business intelligence; statistical analysis; relationship scoring; customer segmentation.

Executive summary for government stakeholders

1. Purpose and public-sector relevance

This research examines how recorded banking relationships differ between retained and churned customers. For public-sector stakeholders, its value lies in illustrating evidence-based analytical practice, not estimating a national or European market problem. The sample's bank, sampling design and reporting period are not established. It cannot represent all European banks, public institutions or financial consumers. The central distinction is between owning a product and using it: neither product count nor card ownership proves meaningful engagement.

2. Dataset and analytical approach

The available charts describe 10,000 observations and contain activity, product-count, balance, estimated-salary, card-ownership, geography, age-band, tenure-band, credit-score-band and churn information. Reporting year and original variable count cannot be verified. Twelve analytical outputs and six supplied dashboard screenshots were reviewed. Counts were reconciled, and supplementary confidence intervals and association tests were calculated from the embedded data. The dashboard illustrates filtered population review. This is observational research; it does not show what would happen if a bank changed a customer's activity or product portfolio.

3. Key findings

Overall, 2,037 observations were classified as churned (20.37%) and 7,963 as retained (79.63%). Among 4,849 inactive customers, 26.85% churned, compared with 14.27% of 5,151 active customers. The 12.58-percentage-point difference identifies a group worth investigating, not an expected benefit from contacting it. Product-count results are strongly non-linear: churn was 27.71% for one product, 7.58% for two, 82.71% for three and 100.00% for four. The last two groups contained only 266 and 60 observations. They require source validation before operational conclusions.

A reconstructed 80th-percentile balance threshold of 133,710.358, in unspecified currency units, identifies 2,000 high-balance customers. Of these, 985 were inactive, and 296 churned (30.05%). Cardholder retention was 79.82%, versus 79.19% without a card; this small difference is statistically inconclusive. Reported relationship-score bands show decreasing churn from 34.54% to 12.95%, but the score is heuristic rather than a validated probability.

4. Implications for government and public-sector stakeholders

These findings support investment in transparent definitions, reproducible reporting and responsible analytical capabilities. They do not establish a need for a specific regulation, marketing program or public subsidy. Product accumulation should not be adopted as a policy success measure: two-product customers had low churn, but customers with more products did not. Any policy hypothesis concerning engagement or access requires representative data, outcome timing and separate evaluation.

5. Recommended actions

First, improve actual-use measurement because current indicators describe holdings and activity status rather than transaction frequency; the intended outcome is better segment definition, evaluated against longitudinal behavior. Second, validate unusually high three- and four-product churn through a customer-level source audit before outreach. Third, review inactive high-balance relationships through consent-aware service conversations, testing whether the identified concerns are genuine. Fourth, evaluate proposed retention actions with randomized or otherwise credible comparison groups, using subsequent churn, complaints and contact burden. Each action is a testable recommendation, not a promised retention gain.

6. Responsible data use, privacy, and limitations

Customer identifiers and surnames are unnecessary for population-level analysis. Minimize data collection, separate identifiers from analytical extracts, restrict access and record permitted uses. Demographic comparisons should prompt fairness and service-quality review, not automatic exclusion. Published HTML charts can themselves contain individual financial observations and therefore need privacy review. The score must not become an unvalidated automated eligibility rule. Unknown sampling, unavailable raw-data checks, limited behavioral detail, small segments and uncertain reporting dates constrain the findings. A constant year, if later confirmed, would preclude year-over-year analysis.

7. Decisions and future data requirements

Prioritize a complete source audit, richer engagement measurement and controlled pilots rather than deployment of a churn-targeting rule. Transaction histories, product-level usage, service contacts, explicit churn dates and experimentally measured intervention outcomes are the highest-value additions. Transparent analytics can improve the questions stakeholders ask; only validation and ongoing evaluation can establish whether resulting decisions are effective and appropriate.

Table of contents

Abstract 1
Executive summary for government stakeholders 2
1. Purpose and public-sector relevance 2
2. Dataset and analytical approach 2
3. Key findings 2
4. Implications for government and public-sector stakeholders 2
5. Recommended actions 3
6. Responsible data use, privacy, and limitations 3
7. Decisions and future data requirements 3
List of figures 6
List of tables 7
Chapter 1: Introduction 8
Chapter 2: Banking Customer Retention Problem Statement 8
Chapter 3: Research Objectives and Questions 9
Chapter 4: Literature Review and Conceptual Background 10
4.1 Engagement and relationship management 10
4.2 Segmentation, analytics and business intelligence 10
4.3 Statistical evidence, prediction and economic decisions 10
4.4 Intervention evaluation and research gap 11
6.2 Measures and uncertainty 12
6.3 Association tests and interpretation 13
6.4 Tools actually used and reconstruction boundaries 13
Chapter 7: Data Cleaning and Validation 14
Chapter 8: Exploratory Data Analysis and Overall Churn Profile 15
Chapter 9: Customer Engagement and Churn Analysis 16
Chapter 10:Product Count and Retention Analysis 17
Chapter 11: Financial Commitment Versus Customer Engagement 19
11.1 Balance distribution and outcome differences 19
11.2 High-balance activity comparison 20
11.3 Salary and balance are not interchangeable 21
Chapter 12: Credit Card Ownership and Retention 22
Chapter 13: Supporting Customer Characteristics 23
13.1 Geography 23
13.2 Age 24
13.3 Tenure, credit score and unavailable gender analysis 24
Chapter 14: Customer Segmentation and Retention-Priority Classification 25

14.1 Hierarchy and overlap audit 26
Chapter 15: Key Performance Indicators and Relationship Strength Index 26
15.1 Operational definitions and denominators 26
15.2 Score design versus verified score behavior 27
Chapter 16: Results and Statistical Interpretation 28
16.1 Supplementary association tests 29
Chapter 17: Streamlit Dashboard Design and Implementation 29
17.1 Verified overview filter states 30
17.2 Chart interpretation and implementation limits 32
Chapter 18: Business Recommendations 33
Chapter 19: Limitations 34
19.1 Data provenance and measurement 34
19.2 Design, uncertainty and inference 34
19.3 Scores, dashboard and external validity 34
Chapter 20: Future Research and Additional Data Requirements 35
20.1 Build an event-time customer evidence base 35
20.2 Prediction and score validation 35
20.3 Establish intervention effectiveness separately 35
Chapter 21: Conclusion 36
References 37
Primary project evidence 37
Appendix A: Additional analytical graphs 38
B.1 Principal contrasts and denominator checks 39
B.2 Supporting-variable outcome counts and intervals 39
Appendix D: Methodology and reproducibility 43
D.1 Reproducing the available-evidence analysis 43
D.2 Evidence-to-calculation manifest 43
D.3 Outstanding verification and missing requested assets 44

List of figures

Figure 1. Overall observed outcome distribution (n = 10,000)
Figure 2. Observed churn by recorded activity status
Figure 3. Observed churn by individual product count
Figure 4. Customer counts by product count and churn outcome
Figure 5. Balance distributions by activity and churn outcome
Figure 6. Estimated salary versus balance in the displayed 3,000-customer sample
Figure 7. Observed churn by credit-card ownership
Figure 8. Observed churn by recorded geography
Figure 9. Observed churn by age band
Figure 10. Observed churn across the supplied retention-review groups
Figure 11. Observed churn by Relationship Strength Index band
Figure 12. Supplied dashboard overview: all activity statuses
Figure 13. Supplied dashboard overview: active customers
Figure 14. Supplied dashboard overview: inactive customers
Figure 15. Observed churn by tenure band
Figure 16. Observed churn by credit-score band
Figure 17. Supplied dashboard chart excerpt: all-customer state
Figure 18. Supplied dashboard chart excerpt: active-customer state
Figure 19. Supplied dashboard chart excerpt: inactive-customer state

List of tables

Table 1. Research at a glance
Table 2. Available evidence and audit status
Table 3. Expected schema and evidence-qualified data dictionary
Table 4. Validation performed and checks still outstanding
Table 5. Overall churn and retention in the supplied outputs
Table 6. Activity status: customer counts, outcomes and uncertainty
Table 7. Individual product counts: outcomes and Wilson intervals
Table 8. Balance summaries by activity and observed outcome
Table 9. High-balance customers using reconstructed population 80th percentile
Table 10. Card ownership: customer counts, outcomes and uncertainty
Table 11. Supporting characteristics: segment sizes and observed churn
Table 12. Reported retention-review groups: size, churn and financial profile
Table 13. Research KPI definitions and evidence status
Table 14. Relationship-strength bands: observed outcomes and uncertainty
Table 15. Research-question synthesis: findings, evidence and interpretation
Table 16. Newly computed omnibus tests against churn
Table 17. Visible dashboard navigation and audit boundaries
Table 18. Evidence-linked retention recommendations and measurement plan
Table 19. Supplementary contrasts and denominator definitions
Table 20. Supporting partitions: recovered outcomes and Wilson uncertainty
Table 21. Reproduction manifest for the supplied output files
Table 22. Items not supplied or not verifiable in this edition

Figure and table lists are maintained to match this edition. The table of contents is a Word field; update all fields in Word after changing pagination.

Chapter 1: Introduction

Customer attrition is a relationship-management problem as well as a reporting outcome. A bank can observe balances, credit scores and product holdings while knowing comparatively little about whether a customer receives continuing value from the relationship. A positive balance may coexist with inactivity; several products may coexist with dissatisfaction or a planned exit. Retention strategy therefore requires multiple dimensions of evidence rather than a single proxy for loyalty.

This study examines recorded engagement, relationship breadth and financial commitment in supplied customer-churn outputs. The analytical emphasis is on customer activity, one- through four-product relationships, high-balance inactivity and credit-card ownership. Geography, age, tenure and credit score provide supporting context rather than a substitute for behavioral measurement. Relationship-strength bands and rule-based review segments are assessed against observed churn, with particular attention to the limitations of heuristics.

The contribution is an evidence-linked interpretation of the available analysis: original chart values are retained, cross-output consistency is checked, and statistical extensions are explicitly labeled. The Streamlit screenshots show how filtered analytics can support review, while the paper separates visible interface behavior from implementation details that cannot be inspected. The research does not claim market representativeness, identify a causal mechanism or estimate the benefit of a retention campaign. Its practical purpose is to support better-framed investigations and defensible decisions.

Chapter 2: Banking Customer Retention Problem Statement

Churn is operationalized as the recorded binary outcome labeled “Churned” in the chart exports, consistent with the brief’s proposed Exited definition. Retention is its complement within the same eligible observations. Without contractual definitions and dates, it is not possible to establish whether churn means account closure, relationship termination or another event. Consequently, these rates are cross-sectional outcome proportions, not a measured annual hazard.

Attrition can involve the loss of a customer relationship, reduced opportunities to deliver relevant services and acquisition costs for replacement relationships. Those consequences motivate retention analysis but are not monetized here: revenue, cost-to-serve and lifetime-value measures are absent. Treating every churned relationship as equally valuable would be unsupported.

The constructs are deliberately distinct. Activity is a binary status, not an interaction count. Product count measures relationship breadth, not depth of use. Card ownership is a holding indicator, not evidence of spending. Balance is recorded financial stock, whereas salary is an estimated flow. Actual utilization would require product-specific frequency, recency and transaction measures. The problem is therefore to identify where these partial indicators signal further review without mistaking them for customer intent.

Chapter 3: Research Objectives and Questions

The overall objective is to describe churn patterns accurately and translate them into testable retention-review priorities. The questions are addressed through the methods and evidence shown below, rather than through an unimplemented predictive model.

- Measure total churn and retention, with an explicit denominator (Chapters 6 and 8).
- Compare activity groups using counts, uncertainty, absolute differences and rate ratios (Chapter 9).
- Assess all four product-count groups and whether their relationship with churn is linear (Chapter 10).
- Investigate balances, estimated salary and high-balance inactivity using the available plotted observations (Chapter 11).
- Compare card ownership without equating ownership with utilization (Chapter 12).
- Describe geography, age, tenure and credit score; assess gender only if evidence is available (Chapter 13).
- Examine transparent segments, their possible overlaps and the relationship-score bands (Chapters 14 and 15).
- Distinguish descriptive results, statistical associations and business hypotheses; identify data and experimental requirements (Chapters 16–21).

The ten research questions in the brief map to these objectives: activity; individual product groups; non-linearity; inactive high-balance customers; card ownership; combinations of activity, balance and products; supporting characteristics; transparent segments; relationship-score behavior; and future evaluation. Where the source files cannot establish a rule, reporting period or cross-tabulation, the paper gives the available result and states the remaining requirement.

Chapter 4: Literature Review and Conceptual Background

4.1 Engagement and relationship management

Brodie et al. (2011) frame customer engagement as a broader relational concept arising from interactive experiences, not merely product possession. This conceptual distinction matters here: a binary activity flag is at best a narrow operational proxy. It does not measure emotional commitment, service experience or the quality of digital interactions. The study therefore uses “recorded activity” for empirical comparisons and reserves “engagement” for the wider construct.

Verhoef (2003) examines customer relationship-management efforts in relation to retention and customer-share development. That work supports distinguishing different relationship outcomes and mechanisms rather than assuming a universal effect of relationship programs. In the present study, product breadth and observed retention are measured separately. The low churn of two-product customers cannot establish that acquiring a second product would cause lower churn for otherwise comparable customers.

4.2 Segmentation, analytics and business intelligence

Ngai et al. (2009) review applications of data-mining techniques to customer relationship management and provide a useful organizing framework for analytical tasks. Segmentation can summarize heterogeneous relationships and make review rules understandable. It does not, by itself, validate a decision policy. The supplied retention-priority groups are therefore treated as interpretable classifications whose definitions and assignment order require source documentation.

Business intelligence contributes a different function from modeling: it makes denominators, filters and segment outcomes available to users. A dashboard is analytically useful when users can identify the population currently in view and compare like with like. Filtered screenshots also create communication risks if a selected denominator or an off-screen category is overlooked. This motivates the screenshot audit in Chapter 17.

4.3 Statistical evidence, prediction and economic decisions

Neslin et al. (2006) investigate the predictive accuracy of customer-churn models, emphasizing that predictive performance is an empirical question. No comparable model-training or out-of-sample evaluation is available in this project. A descriptive score with sensible weights cannot be presented as equivalent to such a model.

Verbeke et al. (2012), in a telecommunications setting, illustrate a profit-driven approach to churn-model evaluation. The cross-industry lesson is methodological: classification quality and economic decision value are not interchangeable. Their sector-specific results are not assumed to transfer to banking. Without margins, intervention costs and incremental response in this dataset, the economic value of targeting cannot be estimated.

Wilson (1927) supplies the basis for score intervals for proportions. Such intervals are particularly helpful for small groups or extreme observed rates: 60 churns out of 60 observations need not mean that the corresponding population rate is exactly 100%. Intervals describe model-based sampling uncertainty, not bias caused by an unknown sampling design.

4.4 Intervention evaluation and research gap

Ascarza (2018) demonstrates why customers with high churn risk are not necessarily the best targets for retention interventions. Intervention responsiveness is distinct from outcome risk. This supports controlled testing rather than acting on a high observed segment rate alone.

The present study addresses a practical gap between descriptive outputs and defensible retention recommendations. It combines explicit denominator reporting, separate product-count analysis, financial–activity comparisons, score-band interpretation and an audit of dashboard evidence. Its contribution is not a new causal theory or predictive algorithm. It is a transparent assessment of what the supplied banking outputs establish and what must be validated before action.

Chapter 5: Dataset Description and Data Dictionary

The available evidence comprises 12 self-contained Plotly HTML files, six dashboard screenshots and the research brief. Nine categorical-rate outputs each account for 10,000 customers. The balance figure contains 10,000 paired activity–balance observations partitioned by churn outcome; the product histogram contains 10,000 product counts partitioned by outcome. The salary scatter contains 3,000 displayed observations only. These different evidence layers must not be joined row-by-row without identifiers or a documented common ordering.

Table 2: Available evidence and audit status

Evidence	Available content	Verification boundary
12 Plotly HTML outputs	Embedded data, layouts and labels	All 12 files parsed in full; not an audited raw CSV
6 dashboard screenshots	All / active / inactive states and chart excerpts	Authentic supplied images; no live application inspection
Raw customer CSV	Not attached	Original schema, identifiers and missingness unavailable
Notebook / ZIP / app source	Not attached	Cleaning logic, score rules, sampling and downloads unverified

The brief expects 14 variables. The following dictionary distinguishes expected source fields from constructs visible in the outputs. It is not a confirmation that the original CSV has exactly 14 columns or the proposed storage types. Currency is unspecified. CustomerId and Surname should be excluded from analytical predictors and retained separately only when an authorized operational linkage is necessary.

Table 3: Expected schema and evidence-qualified data dictionary

Field	Meaning / expected type	Analytical role and limitation
Year	Record year; integer expected	Unavailable; reporting period and constancy unverified
CustomerId	Customer identifier; ID	Unavailable; uniqueness cannot be tested
Surname	Personal name; text	Unavailable; exclude from predictors
CreditScore	Recorded score; numeric	Bands visible; raw range / provider unknown
Geography	Recorded geography; categorical	France, Germany, Spain visible; representativeness unknown
Gender	Recorded category	No supplied output; analysis not performed
Age	Age; numeric	Bands visible; raw minimum and maximum unverified
Tenure	Years with bank; numeric	Bands 0–2, 3–5, 6–8, 9–10 visible
Balance	Account balance; numeric	10,000 plotted values; currency and observation date unknown
NumOfProducts	Count of associated products; integer	One through four visible; product types unavailable
HasCrCard	Card ownership; binary expected	Cardholder / no-card groups visible; not utilization
IsActiveMember	Activity; binary expected	0 / 1 in plots; operational definition unknown
EstimatedSalary	Estimated salary; numeric	3,000 scatter values; estimation process unknown
Exited	Churn; binary expected	Charts label 0 retained, 1 churned; event timing unknown

Source: expected schema from supplied brief; evidence qualifications based on actual attached outputs. Raw data types are proposed, not inspected.

Chapter 6: Research Methodology

6.1 Design and evidence workflow

The design is quantitative, observational and cross-sectional with respect to the supplied outcomes. The workflow was: inventory every attachment; parse each complete HTML file; decode the Plotly.newPlot data and layout arguments; recover segment counts and rates; verify count arithmetic; compare independent chart totals and screenshot KPIs; calculate explicitly labeled supplementary statistics; render static figures; and interpret findings alongside the literature.

For rate outputs, event counts were recovered as the integer nearest to segment size multiplied by the unrounded embedded rate. Every recovered integer reproduced its source rate within 10^{-10}, and each categorical partition totaled 10,000. The product histogram independently supplied outcome-specific product counts. The balance box plot supplied paired balances and activity for all 10,000 plotted observations. No customer identifiers were reconstructed and no linkage between unrelated trace orderings was assumed.

6.2 Measures and uncertainty

$$\text{Churn rate} = 100 \times \frac{\text{Churned observations}}{\text{Eligible observations}}$$

$$\text{Retention rate} = 100 \times \frac{\text{Retained observations}}{\text{Eligible observations}}$$

The denominator is the population represented by a chart or selected dashboard state, not a newly defined exposure period. Overall eligibility follows the supplied output universe; source inclusion and exclusion rules are unknown. Within each segment, the denominator is its own customer count. Wilson 95% confidence intervals were newly computed from recovered counts (Wilson, 1927). Embedded intervals in the activity and product charts agree to reporting precision; minor differences follow the normal quantile implementation.

The activity contrast uses an unpooled normal interval for the difference between independent proportions and a log-scale interval for the risk ratio. The card retention difference uses the same independent-proportion normal approximation. These intervals are unadjusted descriptive contrasts. A 95% confidence interval does not quantify uncertainty from measurement error, selection bias or unknown population coverage.

6.3 Association tests and interpretation

Nine Pearson chi-square tests without continuity correction were newly performed: activity, product count, card ownership, geography, age band, tenure band, credit-score band, relationship-score band and retention-review group against churn. For each, the null hypothesis is independence between the named category and churn; the alternative is an association in at least one category. The nominal significance level is .05. Cramér's V summarizes unadjusted association strength. All expected counts exceeded five; Chapter 16 reports statistics, degrees of freedom, p-values and effect sizes.

Holm adjustment covers the nine declared omnibus tests; it does not correct all possible exploratory questions. A supplementary two-sided Fisher exact test compares three- and four-product groups because one observed cell is zero. Its p-value is exploratory and outside the nine-test family. The null is equal churn odds; the alternative is unequal odds. The raw odds ratio is zero in the chosen three-versus-four orientation and is not interpreted as a stable effect estimate. No causal regression, multivariable adjustment, predictive model or formal correlation analysis is claimed.

6.4 Tools actually used and reconstruction boundaries

The supplied exports identify Plotly.js version 2.35.2; the screenshots identify a Streamlit-style application but do not prove its internal library stack. For this paper, Python, JSON parsing, NumPy, Pandas, SciPy and Statsmodels were used for extraction and calculations, Plotly and Kaleido for static chart export, and gendocx for the Word document. A Chromium renderer was used for chart images. These are paper-production tools, not an assertion that the unavailable notebook used the same libraries.

The high-balance threshold was newly reconstructed with the standard linear-interpolation 80th percentile over all 10,000 embedded balance values and an inclusive greater-than-or-equal comparison. It reproduces the supplied high-balance segment count and churn exactly. This supports consistency with that definition but does not prove the original configuration. Methods specified only in the brief,

such as exact segmentation precedence or score normalization, are not presented as code-audited implementation.

Chapter 7: Data Cleaning and Validation

No raw records were deleted or imputed for the paper. The analysis uses the observations already present in the supplied chart exports. The term “valid customer records” appears in screenshots, but it is a dashboard label, not independent evidence that every field passed substantive validation. Claims in the brief about zero missing values and zero exact duplicate rows cannot be confirmed without the CSV.

Table 4: Validation performed and checks still outstanding

Check	Result	Interpretation
Categorical partition sizes	All nine sum to 10,000	Consistent output universe
Churn / retention arithmetic	2,037 / 7,963 across output partitions	Internal consistency, not customer-level audit
Recovered integer events	Source rates reproduced within 10^{-10}	Counts supported by embedded rates
Balance / activity trace lengths	10,000 paired observations; activity 0 / 1	Financial summaries reproducible within export
Product outcome histogram	Counts agree with product-rate output	Independent chart cross-check
Balance range / zero values	0 to 250,898.09; 3,617 zeros	Zero retained as observed value, not imputed
Raw missingness / duplicates / IDs	Not verifiable	CSV needed; no claim of zero defects
Raw ranges / categories / Year	Partially visible only	Full schema and validation report needed

Zero balances are legitimate observed values unless a source specification shows otherwise; replacing them with missing values would change the financial distribution and risk introducing bias. Extremes such as four-product churn of 100% are not automatically invalid, but warrant confirmation of target coding, extraction filters, temporal ordering and source provenance. Source field definitions should be obtained before deciding whether unexpected records require correction or exclusion.

A complete future validation pass should check required columns, types, missing values by field, exact duplicates, duplicate customer IDs, allowed categorical and binary values, plausible numeric ranges and churn timestamps. It should preserve a dated audit trail of any exclusions. Year-over-year analysis is not performed: the existence or constancy of Year cannot be established from the attachments.

Chapter 8: Exploratory Data Analysis and Overall Churn Profile

The common output universe contains 10,000 customer observations. There are 2,037 churned observations and 7,963 retained observations, corresponding to 20.37% and 79.63%, respectively. This total is independently supported by the balance plot's outcome traces, the product histogram and all categorical partitions. The full-population dashboard screenshot also displays 10,000 customers, 20.37% churn and 79.63% retention.

Table 5 and Figure 1 summarize the outcome distribution. Retained observations outnumber churned observations by approximately 3.91 to one. A classification system that always predicted retention would therefore achieve 79.63% apparent accuracy, illustrating why accuracy alone would be inadequate if predictive modeling were later attempted. No such model is fitted here.

Table 5: Overall churn and retention in the supplied outputs

Outcome	Customers	Share
Churned	2,037	20.37%
Retained	7,963	79.63%
Total	10,000	100.00%

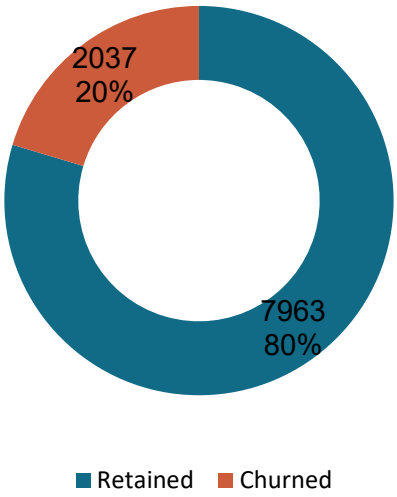


Figure 1: Overall observed outcome distribution (n = 10,000)

Newly generated native Word chart from counts independently reconciled across the supplied HTML exports; not from an independently audited raw CSV.

The newly computed Wilson 95% interval for overall churn is 19.59–21.17%. This interval is conditional on binomial sampling assumptions and does not make the sample representative. Because observation dates and the event window are unknown, neither annualized churn nor survival probability can be inferred. The paper uses this overall rate as a descriptive benchmark for segment comparisons, not a forecast.

Chapter 9: Customer Engagement and Churn Analysis

Table 6: Activity status: customer counts, outcomes and uncertainty

Group	Customers	Exited	Retained	Churn	Retention	95% CI: churn
Inactive	4,849	1,302	3,547	26.85%	73.15%	25.62–28.12%
Active	5,151	735	4,416	14.27%	85.73%	13.34–15.25%

Inactive customers account for 4,849 observations (48.49% of the population) and 1,302 churns. Active customers account for 5,151 observations and 735 churns. Churn is 26.85% among inactive customers compared with 14.27% among active customers. Retention is correspondingly 73.15% and 85.73%. Figure 2 displays both categories with separate denominators, avoiding the ambiguity of a filtered dashboard excerpt that shows only one activity status.

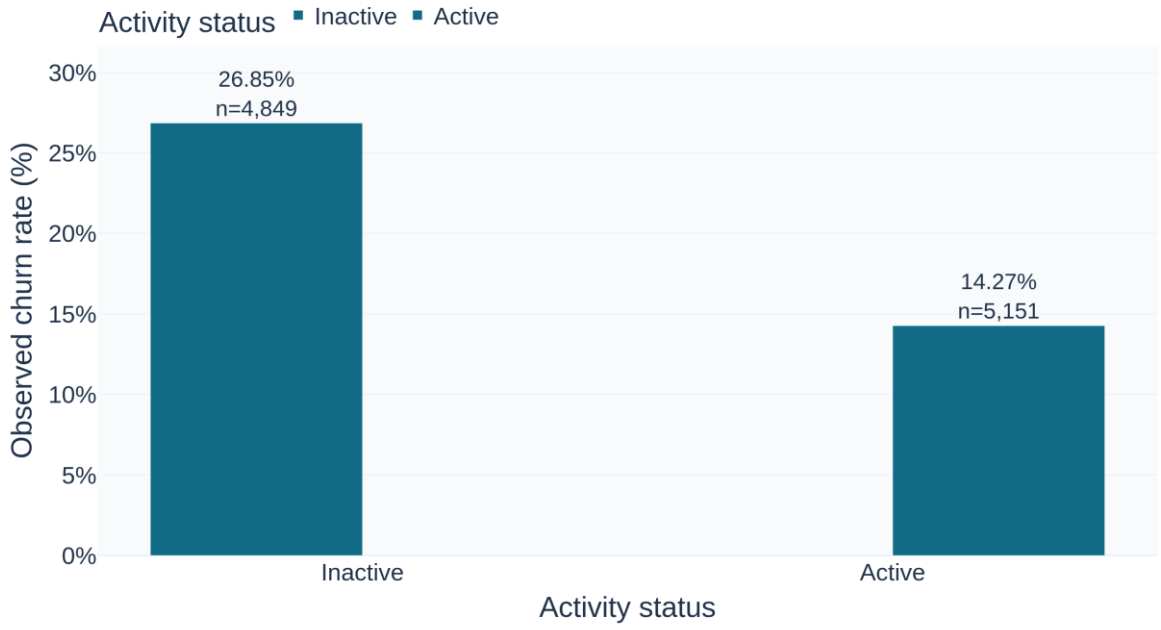


Figure 2: Observed churn by recorded activity status

Source: supplied churn_by_activity.html; embedded Plotly data rendered at 3,000 × 1,700 pixels. Visual formatting standardized; observations unchanged.

The inactive-minus-active churn difference is 12.58 percentage points, with an unadjusted approximate 95% interval of 11.01–14.15 points. The inactive-to-active churn-rate ratio is 1.88 (95% interval 1.73–2.04). These estimates distinguish an absolute gap from a relative comparison; neither is an intervention effect.

The newly calculated Pearson test gives $\chi^2(1) = 243.76$, $p < .001$, with Cramér's $V = .156$; the association remains detectable after Holm adjustment. This is evidence of an unadjusted relationship, not that activity is the strongest predictor. Differences in age, geography, product mix or prior service

experience could contribute. The activity definition may also capture behavior close to exit, making temporal ordering essential.

A practical response is to investigate the reasons behind recorded inactivity and design a service-oriented trial. Reactivation rate, later transactions, complaints, contact burden and subsequent churn should be measured. Simply changing a status label or increasing contact frequency would not establish genuine engagement.

Interpretation: recorded inactivity is an investigation signal; the observed 12.58-point gap is not a promised gain from outreach.

Chapter 10: Product Count and Retention Analysis

Table 7: Individual product counts: outcomes and Wilson intervals

Group	Customers	Exited	Retained	Churn	Retention	95% CI: churn
1	5,084	1,409	3,675	27.71%	72.29%	26.50–28.96%
2	4,590	348	4,242	7.58%	92.42%	6.85–8.38%
3	266	220	46	82.71%	17.29%	77.70–86.78%
4	60	60	0	100.00%	0.00%	93.98–100.00%

The product-count pattern is distinctly non-linear. One-product customers number 5,084, with 1,409 churns (27.71%); two-product customers number 4,590, with 348 churns (7.58%). The one-minus-two churn gap is 20.13 percentage points. Two-product retention is 92.42%, versus 72.29% with one product. This contrast is compatible with relationship differences, but does not identify the effect of cross-selling a second product.

Customers with three products have 220 churns among 266 observations (82.71%). Every one of the 60 four-product observations is labeled churned. The three-product interval is 77.70–86.78%; the four-product interval is 93.98–100.00%. Thus, the observed four-product proportion should not be interpreted as certainty about a wider population. Figure 3 retains the individual categories rather than combining all multi-product customers.

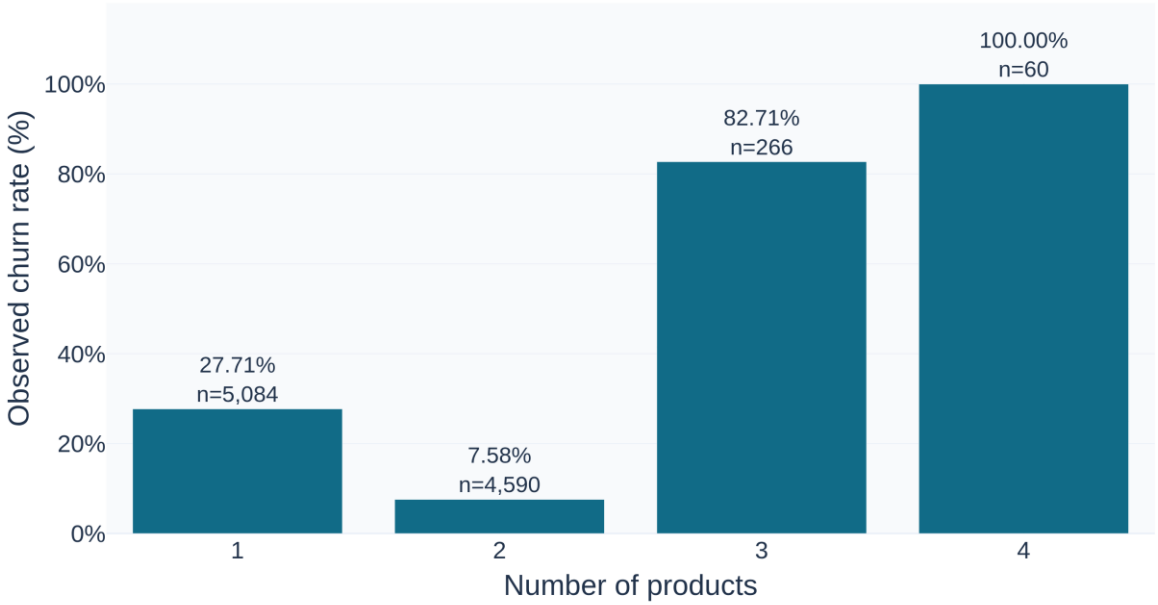


Figure 3: Observed churn by individual product count

Source: supplied churn_by_product_count.html; embedded Plotly data rendered at 3,000 × 1,700 pixels. Visual formatting standardized; observations unchanged.

The product histogram in Figure 4 independently confirms the cell counts: retained customers comprise 3,675 with one product, 4,242 with two, 46 with three and none with four; churned customers comprise 1,409, 348, 220 and 60, respectively. The very small high-product groups are visually dwarfed by the first two categories, which is why the separate rate figure and uncertainty table are needed.

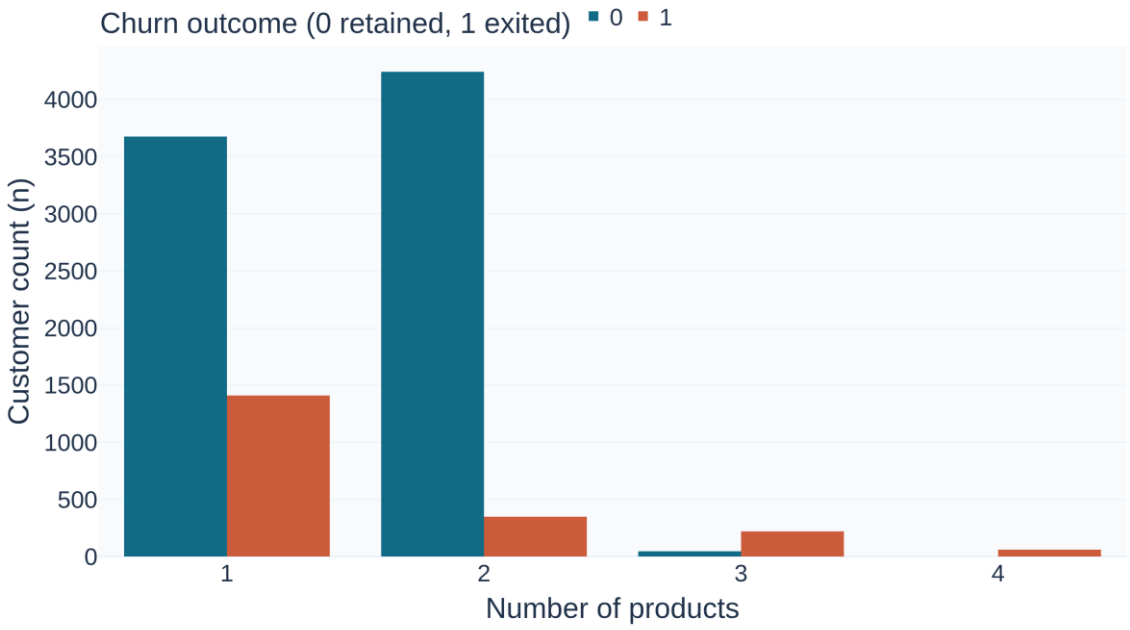


Figure 4: Customer counts by product count and churn outcome

Source: supplied customer_count_by_product_and_churn.html; embedded Plotly data rendered at 3,000 × 1,700 pixels. Visual formatting standardized; observations unchanged.

The omnibus association is substantial in this unadjusted comparison: $\chi^2(3) = 1{,}503.63$, Cramér's V = .388, with p below .001. The numerical p-value underflows to zero in standard double precision; it is reported as $p < .001$ rather than as an exact zero. A supplementary two-sided Fisher comparison of three versus four products gives $p = .0000544$. It is exploratory, was not part of the nine-test Holm family, and does not resolve source validity or confounding.

Possible explanations for high-product churn include specific product combinations, fees, poor fit, pre-exit restructuring or data-generation artifacts. These are hypotheses, not findings. Product types, acquisition dates, usage and closure timing are required to distinguish them. The evidence rejects a simple "more products always means more retention" description of these outputs, but not every alternative explanation. An appropriate first action is source and coding review, followed by product-fit investigation not an automatic prohibition on additional products.

Chapter 11: Financial Commitment Versus Customer Engagement

11.1 Balance distribution and outcome differences

The complete balance export contains 10,000 plotted observations, with a mean balance of 76,485.89, median 97,198.54, minimum 0 and maximum 250,898.09. There are 3,617 zero balances (36.17%). Currency units are unspecified. These descriptive values were newly calculated from the embedded observations; they do not establish account coverage, profitability or financial distress.

Table 8 and Figure 5 compare balance distributions jointly by activity and outcome. Within both activity categories, churned observations have higher average and median balances than retained observations. This directly cautions against interpreting substantial balances as proof of loyalty. It does not establish that higher balances increase churn causally.

Table 8: Balance summaries by activity and observed outcome

Activity / outcome	Customers	Mean balance	Median balance
Inactive / retained	3,547	72,048.82	91,835.16
Inactive / churned	1,302	90,988.81	109,252.73
Active / retained	4,416	73,304.72	92,212.20
Active / churned	735	91,320.64	109,922.61

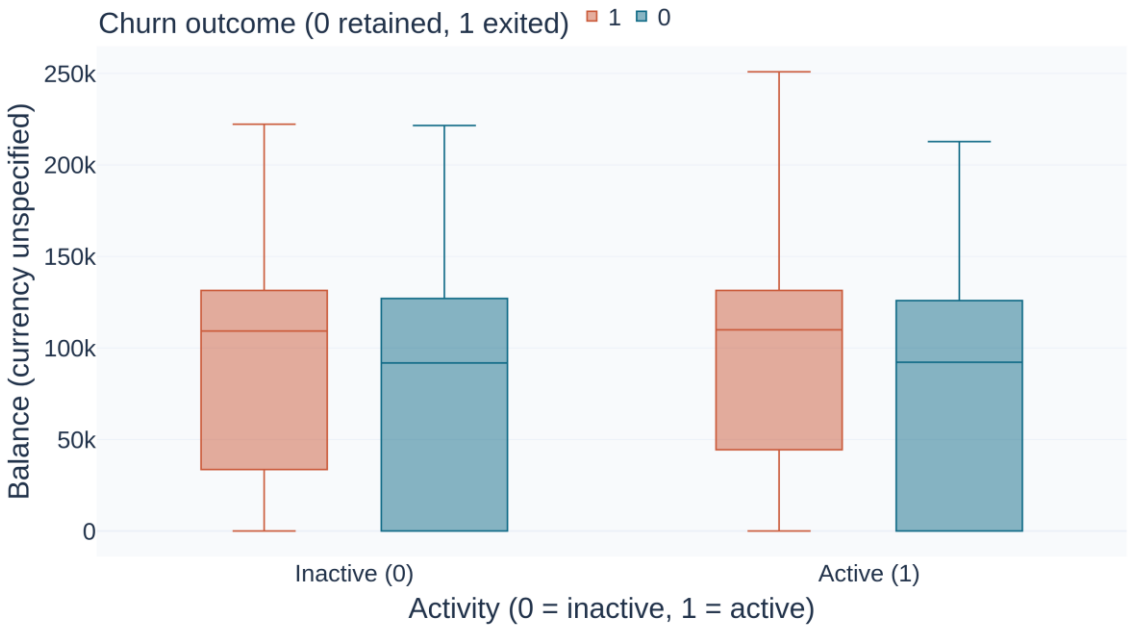


Figure 5: Balance distributions by activity and churn outcome

Source: supplied balance_activity_churn.html; embedded Plotly data rendered at 3,000 × 1,700 pixels. Visual formatting standardized; observations unchanged.

11.2 High-balance activity comparison

The newly reconstructed 80th-percentile threshold is 133,710.358. Applying balance ≥ this unrounded value identifies 2,000 observations. Of these, 985 are inactive, giving a high-balance disengagement share of 985 / 2,000 = 49.25%. This is different from 985 / 10,000 = 9.85%, the share of the entire output population that is both inactive and high balance.

Among the 985 inactive high-balance observations, 296 churned (30.05%). Among 1,015 active high-balance observations, 166 churned (16.35%). The unadjusted gap is 13.70 percentage points. The inactive high-balance count and outcome match the supplied priority segment exactly; the active comparison is a newly derived complement. Agreement supports consistency with the percentile definition, but the absent source code prevents direct verification of its original configuration.

Table 9: High-balance customers using reconstructed population 80th percentile

Activity	Customers	Churned	Churn	95% CI: churn
Inactive	985	296	30.05%	27.27–32.99%
Active	1,015	166	16.35%	14.21–18.76%

New calculations from complete balance–activity–outcome traces. Threshold uses all 10,000 observations, inclusive comparison and full precision; no currency inferred.

11.3 Salary and balance are not interchangeable

Figure 6 displays the 3,000 observations actually embedded in the salary–balance scatter, not the full population. Activity is encoded by marker shape and churn by color. The prominent zero-balance band occurs across a broad salary range. Higher estimated salary does not mechanically imply a high balance in the displayed data; similarly, high balances appear at different salary levels. The chart alone is not a formal test of independence.

High-salary / low-balance and low-salary / high-balance profiles are reasonable review concepts, but the source does not define thresholds or provide full-population joint counts. Their population sizes and churn rates are therefore not reported. The sampling method and random seed for the 3,000-point display are unknown. No salary–balance correlation from that sample is generalized to all customers.

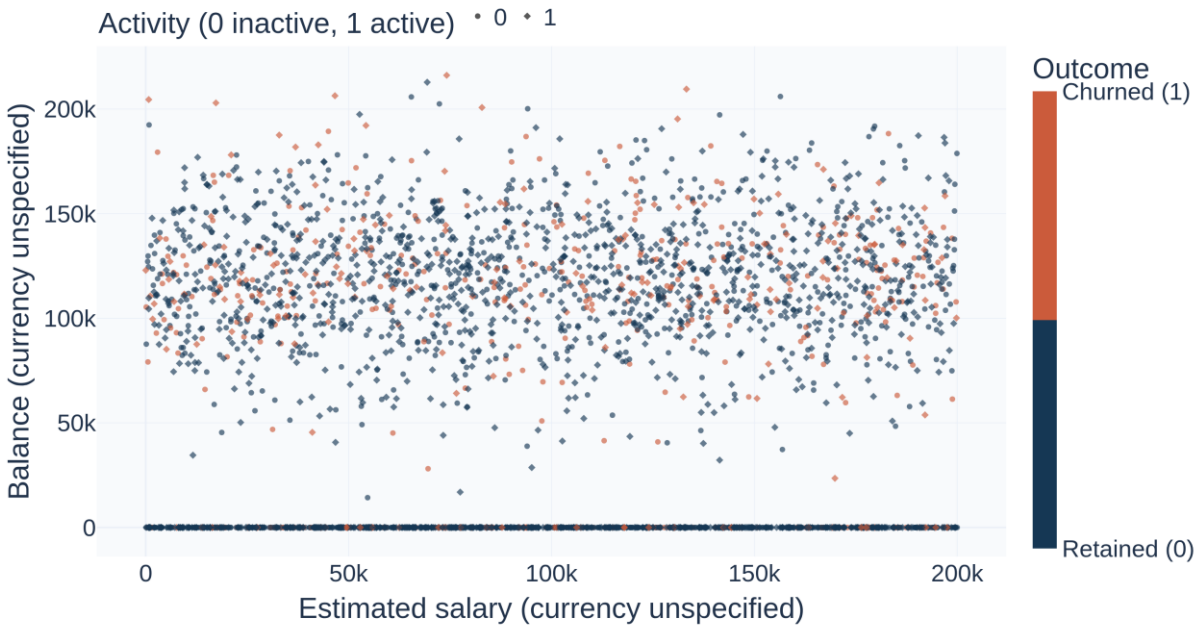


Figure 6: Estimated salary versus balance in the displayed 3,000-customer sample

Source: supplied salary_vs_balance.html; 3,000 embedded observations rendered without changing values. SVG scatter replaces WebGL for reliable static export. Currency unspecified; sample selection is unverified.

Inactive high-balance relationships merit service review because they combine substantial recorded funds and an inactivity flag with an elevated observed churn proportion. However, deposits are not net profit, estimated salary is not verified income, and low balances do not establish hardship. More complete product holdings, fund flows, customer objectives and cost-to-serve data are necessary before assigning economic priority.

Chapter 12: Credit Card Ownership and Retention

Table 10: Card ownership: customer counts, outcomes and uncertainty

Group	Customers	Exited	Retained	Churn	Retention	95% CI: churn
No card	2,945	613	2,332	20.81%	79.19%	19.39–22.32%
Cardholder	7,055	1,424	5,631	20.18%	79.82%	19.26–21.14%

Cardholders number 7,055 and include 1,424 churned observations. Non-cardholders number 2,945 and include 613 churned observations. Churn is 20.18% among cardholders and 20.81% among non-cardholders, while retention is 79.82% and 79.19%. Figure 7 shows a much smaller contrast than the activity or individual product-count comparisons.

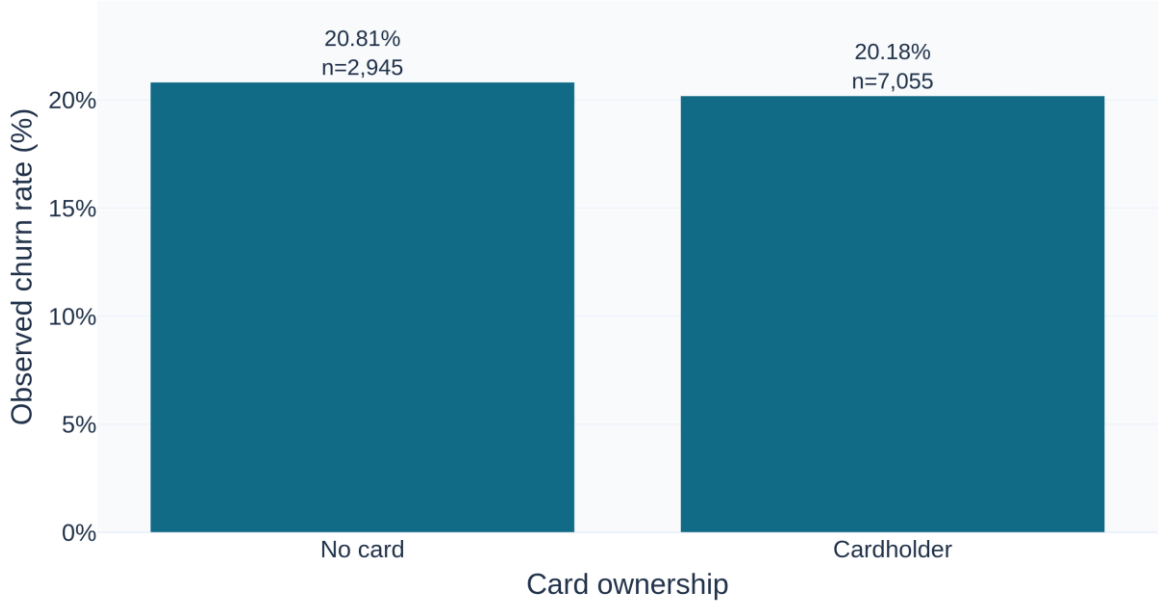


Figure 7: Observed churn by credit-card ownership

Source: supplied churn_by_card_ownership.html; embedded Plotly data rendered at 3,000 × 1,700 pixels. Visual formatting standardized; observations unchanged.

The cardholder-minus-non-cardholder retention difference is 0.63 percentage points. Its approximate 95% interval is -1.11 to 2.37 percentage points, including zero. Pearson $\chi^2(1) = 0.51$, $p = .475$, and Cramér's $V = .007$. These data do not provide persuasive evidence of an ownership-related difference at the declared .05 level. Failure to reject independence is not proof that the true association is exactly zero.

Commercial importance also remains uncertain. Even a precisely estimated small gap would need intervention costs and causal response to justify a campaign. Card ownership cannot measure transaction frequency, revolving behavior, payment regularity or card satisfaction. The appropriate extension is card-usage research, not an assumption that issuing more cards will improve retention.

Chapter 13: Supporting Customer Characteristics

Supporting characteristics contextualize the principal engagement and financial analysis. They are not causal explanations or automatic targeting criteria. Table 11 reports the supplied partition sizes and churn rates; Figure 8 shows geography, Figure 9 age, and Appendix A contains the tenure and credit-score figures. Supplementary outcome counts and confidence intervals are available in Appendix B.

Table 11: Supporting characteristics: segment sizes and observed churn

Characteristic	Group	Customers	Churn
Geography	France	5,014	16.15%
Geography	Germany	2,509	32.44%
Geography	Spain	2,477	16.67%
Age	18–29	1,641	7.56%
Age	30–39	4,346	10.88%
Age	40–49	2,618	30.79%
Age	50–59	869	56.04%
Age	60+	526	27.95%
Tenure (years)	0–2	2,496	21.15%
Tenure (years)	3–5	3,010	20.76%
Tenure (years)	6–8	3,020	18.87%
Tenure (years)	9–10	1,474	21.30%
Credit score	<500	632	23.73%
Credit score	500–599	2,402	21.23%
Credit score	600–699	3,818	19.72%
Credit score	700–799	2,493	19.90%
Credit score	800+	655	19.54%

13.1 Geography

Germany has 814 churns among 2,509 observations (32.44%), compared with 810 of 5,014 in France (16.15%) and 413 of 2,477 in Spain (16.67%). The unadjusted association is detectable: $\chi^2(2) = 301.26$, $p < .001$, $V = .174$. It remains after Holm adjustment. Differences in customer composition, services or sampling could explain part of this pattern; national geography itself is not an identified causal mechanism.

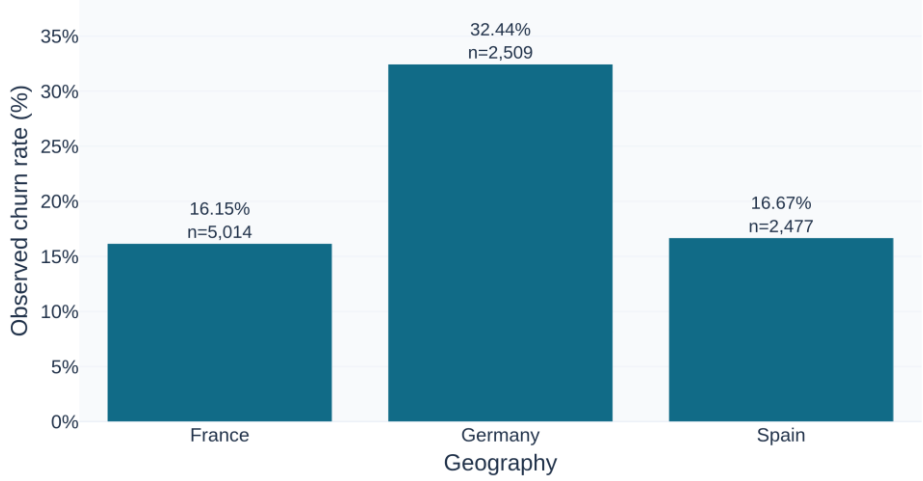


Figure 8: Observed churn by recorded geography

Source: supplied churn_by_geography.html; embedded Plotly data rendered at 3,000 × 1,700 pixels. Visual formatting standardized; observations unchanged.

13.2 Age

Observed churn is 7.56% for ages 18–29, 10.88% for 30–39, 30.79% for 40–49, 56.04% for 50–59 and 27.95% for 60+. The 50–59 band contains 869 observations and 487 churns, so its high proportion is not based on only a few records. Nevertheless, the decline in the 60+ group shows that the relationship is not simply linear across age. Pearson $\chi^2(4) = 1{,}282.67$, $p < .001$, $V = .358$. These are unadjusted associations and do not establish age-related service failures or justify discriminatory treatment.

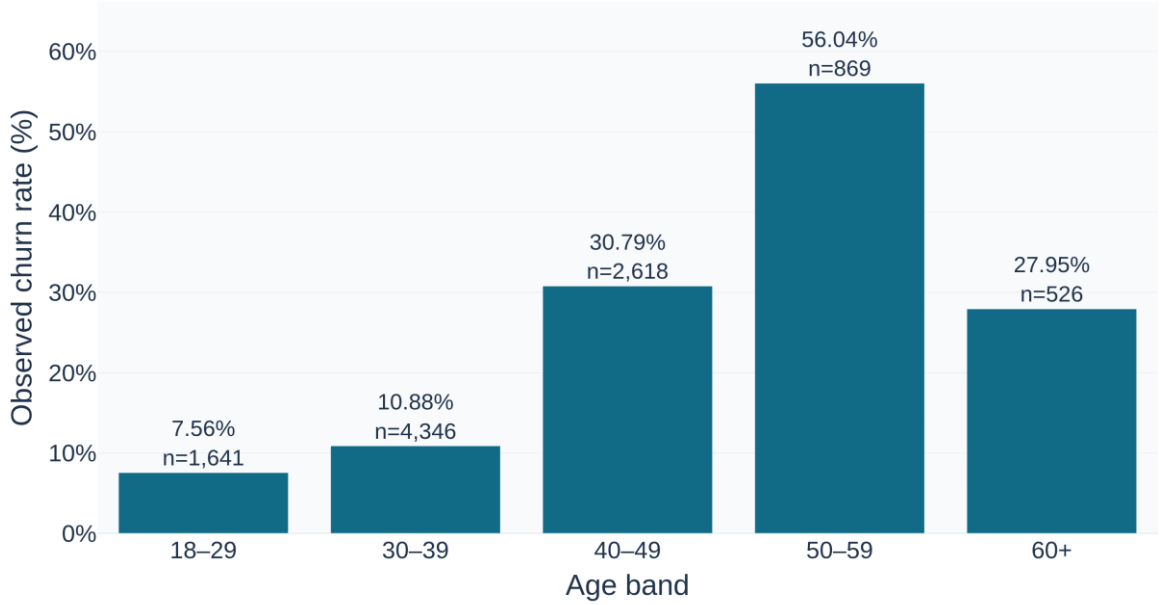


Figure 9: Observed churn by age band

Source: supplied churn_by_age_band.html; embedded Plotly data rendered at 3,000 × 1,700 pixels. Visual formatting standardized; observations unchanged.

13.3 Tenure, credit score and unavailable gender analysis

Tenure-band churn ranges from 18.87% for six to eight years to 21.30% for nine to ten years, with 21.15% for zero to two and 20.76% for three to five years. The omnibus p-value is .103 and V = .025; Holm-adjusted p = .308. Credit-score-band churn ranges from 23.73% below 500 to 19.54% at 800+, but the omnibus p-value is .130 and V = .027; adjusted p = .308. Neither comparison provides strong evidence of association under the declared test family. Absence of a detected band association does not rule out a different relationship in raw continuous values.

No gender analysis is supplied and the raw CSV is unavailable. Gender-related rates are therefore omitted rather than inferred from a familiar dataset. Any future demographic analysis should be framed

around fairness, accessibility and service needs, with appropriate legal and governance review, not automatic exclusion from services.

Chapter 14: Customer Segmentation and Retention-Priority Classification

The supplied retention-priority chart contains five labels, each with a customer count, churn rate, retention rate and average balance. Their counts sum to 10,000 and their recovered churn counts sum to 2,037. This is consistent with a complete assignment into five groups, but aggregate totals alone cannot prove that customer identities are non-overlapping. Exact assignment logic and precedence remain unverified.

Table 12: Reported retention-review groups: size, churn and financial profile

Reported group	n / share	Churn / retention	Mean balance
Review inactive customer	3,718 / 37.18%	23.61% / 76.39%	57,333.21
Opportunity active, one product	2,563 / 25.63%	18.92% / 81.08%	98,902.02
Standard monitoring	2,446 / 24.46%	5.56% / 94.44%	51,852.94
Priority 1 inactive, high balance	985 / 9.85%	30.05% / 69.95%	153,745.86
Review 3 or 4 products	288 / 2.88%	84.03% / 15.97%	69,222.88

The general inactive-review group contains 3,718 customers and 878 churns (23.61%). The active, one-product opportunity group contains 2,563 customers and 485 churns (18.92%). Standard monitoring contains 2,446 customers and 136 churns (5.56%). The inactive high-balance group contains 985 customers and 296 churns (30.05%). The labeled three-or-four-product review group contains 288 customers and 242 churns (84.03%). Figure 10 visualizes these observed rates; “priority” does not mean an intervention has been shown to work.

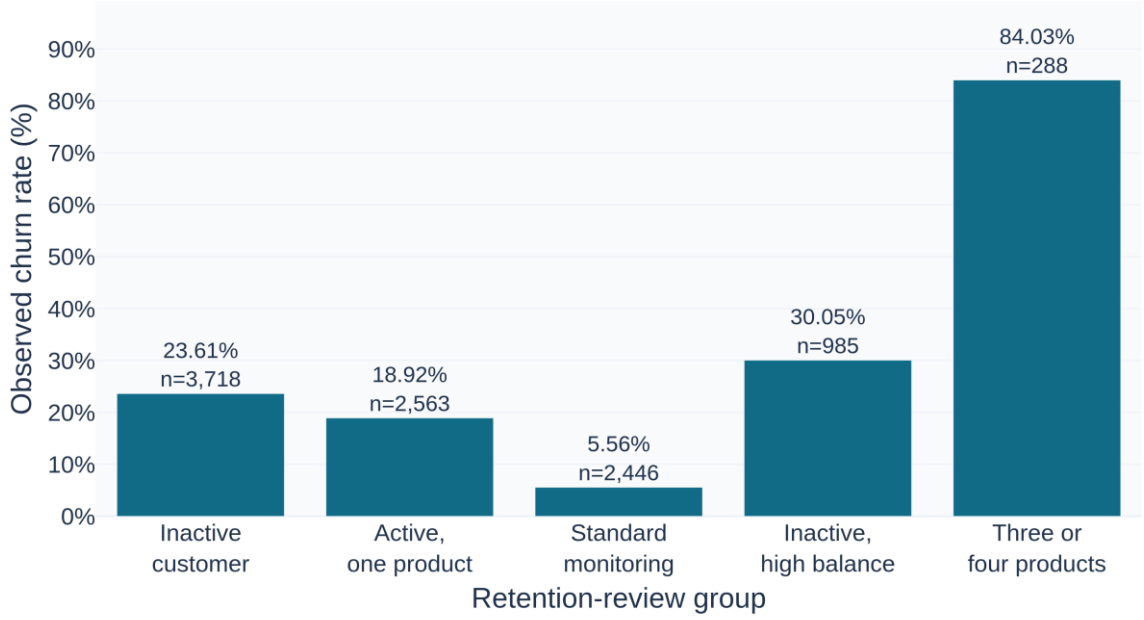


Figure 10: Observed churn across the supplied retention-review groups

Source: supplied retention_priority_by_churn.html; embedded Plotly data rendered at 3,000 × 1,700 pixels. Visual formatting standardized; observations unchanged.

14.1 Hierarchy and overlap audit

The raw three- and four-product categories contain 326 customers, whereas the labeled high-product review group contains 288 a difference of 38. Likewise, all inactive customers total 4,849, but the general inactive group contains only 3,718. The high-balance inactive group accounts for 985 of that difference, leaving 146 observations for other classifications if the labels are exclusive. These identities make category precedence consequential; group names cannot be read as unrestricted marginal definitions.

An assignment order that places high-balance inactivity and selected active profiles ahead of some other rules could explain these differences, but the evidence does not establish the actual order. No such hierarchy is asserted as implemented. Source code or a customer-level assignment file is required to verify every branch. Median balances, within-segment product distributions and complete activity composition are unavailable for the five reported labels.

Changing the balance percentile would change both segment membership and rates; the threshold should be recorded with every comparison. Transparent labels are useful for service-review queues and auditability, but should not be described as customer intent, calibrated risk or an automatic marketing mandate. Reviews should document what is learned, not just the number of customers contacted.

Chapter 15: Key Performance Indicators and Relationship Strength Index

15.1 Operational definitions and denominators

A useful KPI is explicit about its population, numerator, denominator and limitations. The screenshots visibly establish four overview KPIs: customers in view, observed churn, retention and inactive customers. The five analytical KPIs requested in the brief are documented below as research definitions; their complete implementation in unseen dashboard tabs cannot be verified.

Table 13: Research KPI definitions and evidence status

KPI	Definition / result	Evidence qualification
Engagement retention comparison	Active retention 85.73%; inactive 73.15%; churn gap 12.58 points; ratio 1.88	Reconciled across source activity output and screenshots
Product depth / breadth	Mean count 1.5302; one / two retention 72.29% / 92.42%	Mean newly computed from source counts; not utilization
High-balance disengagement	985 inactive / 2,000 high balance = 49.25%	New threshold reconstruction; full population basis
Card ownership retention comparison	79.82% − 79.19% = +0.63 points	Interval includes zero; ownership only

KPI	Definition / result	Evidence qualification
Relationship Strength Index	Requested 50 / 35 / 15 weighting; band results available	Formula supplied in brief; code and score values absent

15.2 Score design versus verified score behavior

The brief specifies RSI = 50 × activity component + 35 × product-breadth component + 15 × card-ownership component, with each component normalized to 0–1. Activity and card ownership are intended as binary components; product credit is capped at two products. The exact normalization of one product, boundary conventions, rounding and missing-value handling cannot be verified without implementation code. The equation below documents the requested design, not a source-code audit.

$$\mathrm{RSI} = 50A + 35P + 15C$$

A = activity component; P = normalized, capped product-breadth component; C = card-ownership component. Specified in the brief; exact implementation unverified.

Table 14: Relationship-strength bands: observed outcomes and uncertainty

Group	Customers	Exited	Retained	Churn	Retention	95% CI: churn
0–25	718	248	470	34.54%	65.46%	31.15–38.09%
26–50	4,131	1,054	3,077	25.51%	74.49%	24.21–26.87%
51–75	788	170	618	21.57%	78.43%	18.84–24.58%
76–100	4,363	565	3,798	12.95%	87.05%	11.99–13.98%

The reported bands contain 718 customers at 0–25, 4,131 at 26–50, 788 at 51–75 and 4,363 at 76–100: shares of 7.18%, 41.31%, 7.88% and 43.63%. Observed churn decreases monotonically across these four aggregate bands: 34.54%, 25.51%, 21.57% and 12.95%. Retention increases from 65.46% to 87.05%. Figure 11 supports an ordered descriptive relationship in this sample; it does not establish a calibrated mapping from score to individual risk.

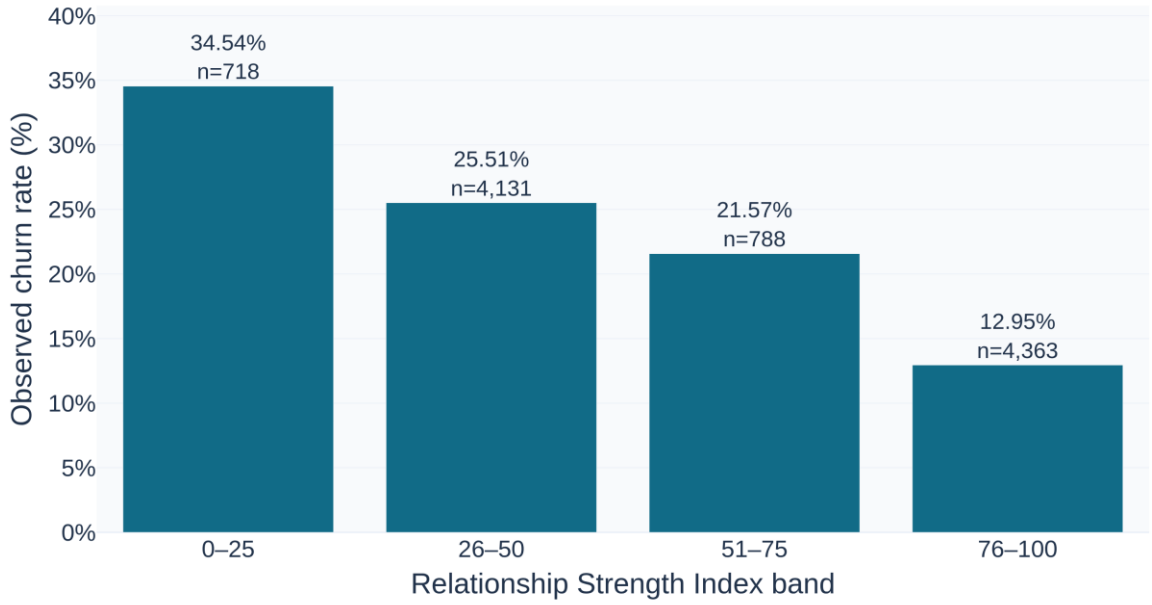


Figure 11: Observed churn by Relationship Strength Index band

Source: supplied relationship_strength_by_churn.html; embedded Plotly data rendered at 3,000 × 1,700 pixels. Visual formatting standardized; observations unchanged.

The newly computed omnibus result is $\chi^2(3) = 305.08$, $p < .001$, $V = .175$. Mean balances by ascending band are 100,598.78; 73,056.08; 100,577.89; and 71,413.92, further illustrating that score and balance are distinct dimensions. No individual score distribution, out-of-time calibration, discrimination, sensitivity analysis or incremental contribution analysis is available.

The score gives no additional product-breadth credit above its two-product cap, even though the raw three- and four-product groups have unusually high observed churn. Aggregated score-band monotonicity can therefore conceal within-band heterogeneity. Weight sensitivity, comparisons against activity alone and separate checks of high-product customers are necessary before deployment. The RSI is a heuristic relationship summary not a machine-learning model or churn probability.

Chapter 16: Results and Statistical Interpretation

The results distinguish what is observed, what formal tests establish under their assumptions, and what decisions might be considered. The synthesis below addresses the ten research questions. Statistical tests are supplementary analyses newly performed for this paper, not results attributed to an unseen notebook.

Table 15: Research-question synthesis: findings, evidence and interpretation

Research question	Observed result	Statistical evidence	Interpretation / limitation
Activity and churn	26.85% inactive vs 14.27% active	V = .156; adjusted p < .001	Review inactivity; confounding and timing unresolved
Individual product groups / non-linearity	27.71%, 7.58%, 82.71%, 100%	V = .388; adjusted p < .001	No universal cross-sell rule; last groups small
Inactive high-balance segment	985 customers; 30.05% churn	Wilson interval 27.27–32.99%	Service-review candidate; not proven profitability
Card ownership	Retention advantage +0.63 points	p = .475; interval −1.11 to +2.37 points	Ownership alone provides little differentiation
Activity × product × balance	Priority labels differ from marginal categories	No full joint table available	Exact hierarchy requires source code
Supporting characteristics	Germany and age 50–59 elevated	Geography / age adjusted p < .001	Unadjusted, not causal or discriminatory rules
Tenure / credit score	Modest band differences	Adjusted p = .308 for both	Do not claim equivalence; banding loses detail
Transparent segments	Five labels; churn 5.56–84.03%	V = .337; adjusted p < .001	Rules interpretable; exclusivity not proved
Relationship score	Churn falls 34.54% to 12.95%	V = .175; adjusted p < .001	Monotonic bands, no individual validation
Future decisions	Source audit and controlled pilots needed	No intervention outcomes available	No causal efficacy or savings estimate

16.1 Supplementary association tests

Table 16: Newly computed omnibus tests against churn

Comparison	χ^2	df	Raw p	Holm p	V	Minimum expected
Activity	243.76	1	<.001	<.001	0.156	987.74
Age band	1282.67	4	<.001	<.001	0.358	107.15
Card ownership	0.51	1	0.475	0.475	0.007	599.90
Credit-score band	7.12	4	0.130	0.308	0.027	128.74
Geography	301.26	2	<.001	<.001	0.174	504.56
Product count	1503.63	3	<.001	<.001	0.388	12.22
Tenure band	6.19	3	0.103	0.308	0.025	300.25
RSI band	305.08	3	<.001	<.001	0.175	146.26
Review group	1134.59	4	<.001	<.001	0.337	58.67

Pearson tests without continuity correction. H_0: category and churn independent; H_1: associated. $\alpha = .05$. Holm family: nine omnibus tests. All expected cell counts > 5. Product p underflows numerically and is shown as <.001, not zero.

Activity, product count, geography, age, review group and RSI band remain associated with churn after the declared multiplicity adjustment. Card ownership, tenure band and credit-score band do not. Cramér's V is an unadjusted association statistic, not model feature importance. Different grouping granularities, overlapping information and sample composition complicate cross-variable comparisons.

The extreme product-count pattern deserves priority verification even though the omnibus test is highly significant. A small p-value cannot confirm that the dataset is correctly labeled, temporally ordered or representative. Equally, an inconclusive ownership contrast does not prove that every usage-based card variable would be uninformative. Commercial significance requires costs, customer outcomes and incremental response, none of which is supplied.

The confidence intervals and tests are conditional on independent observations and an appropriate sampling interpretation. Unknown customer duplication, clustering or synthetic data provenance would affect that interpretation. This is why the paper reports exact output counts while remaining cautious about inference beyond the exported population.

Chapter 17: Streamlit Dashboard Design and Implementation

The screenshots show a dark-themed interface titled "Customer Engagement & Product Utilization Analytics," accompanied by the caution "results are associations, not proof of causation." A sidebar provides optional customer CSV upload, geography selection, activity status, product-count selection and card-ownership selection. The visible uploader accepts CSV and shows a 200 MB limit; parsing, validation, caching and storage behavior are not visible.

The main pane exposes customers in view, observed churn rate, retention rate and inactive customers. A "Churn overview" message explicitly states the filtered population relative to 10,000 records. Tabs visibly read Overview, Engagement, Product Count, Financial Review and Relationship Score; the last

tab is clipped after “Dat” in the images. The six page names requested in the brief are conceptual correspondences, not evidence that six separately inspectable pages were supplied.

Table 17: Visible dashboard navigation and audit boundaries

Visible / requested area	Analytical question	Verified status
Overview / Executive Overview	How many observations and outcomes are in view?	Three overview filter states supplied
Engagement / Engagement Analytics	How does recorded activity relate to churn?	Overview chart excerpts supplied; separate tab not shown
Product Count / Product Utilization	Which product groups differ?	Overview chart excerpts supplied; separate tab not shown
Financial Review / High-Value Detector	Which financial–activity profiles merit review?	Tab visible; contents not supplied
Relationship Score / Scoring	How do score bands relate to outcomes?	Tab visible; contents not supplied
Clipped Dat… / Data Quality and Methodology	What checks and definitions support the analysis?	Label incomplete; page contents unverified

17.1 Verified overview filter states

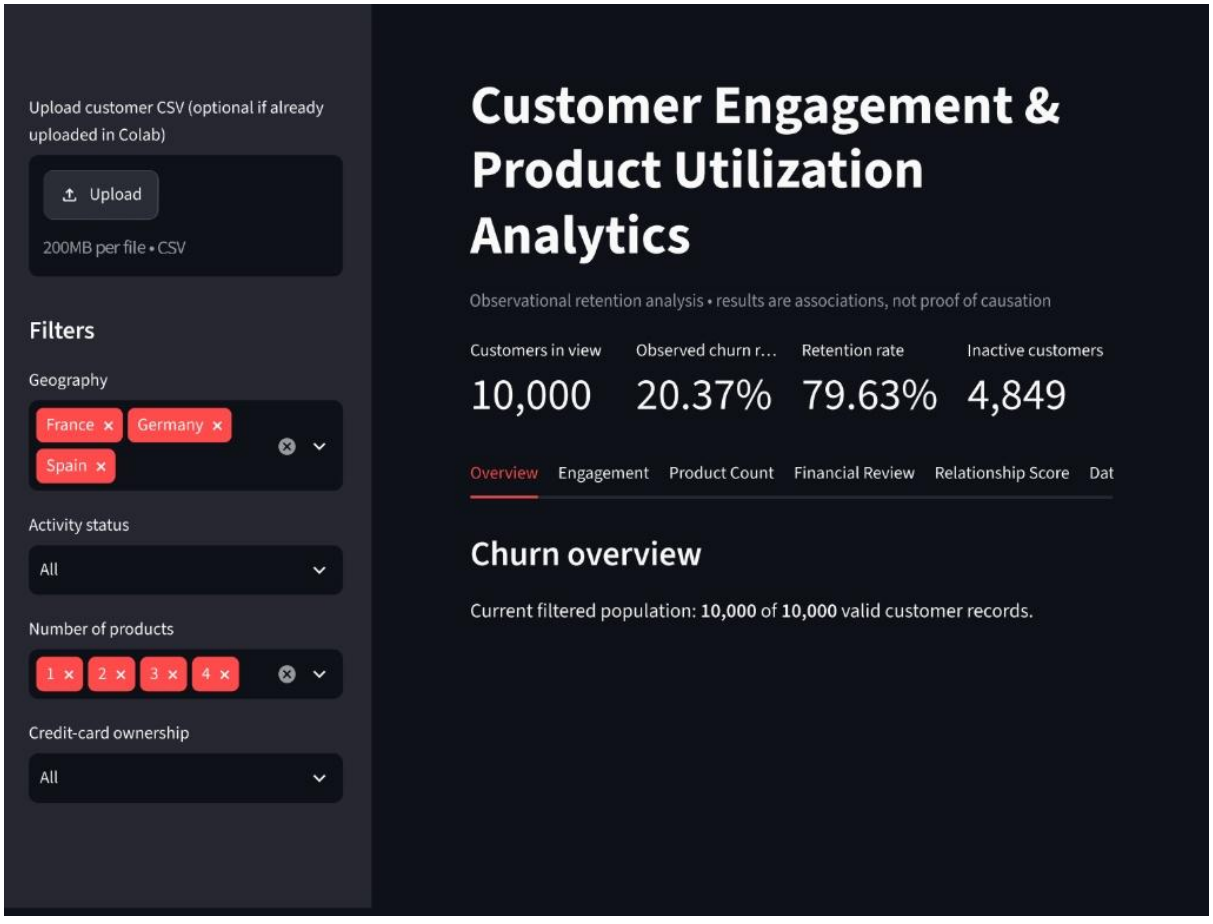


Figure 12: Supplied dashboard overview: all activity statuses

Source: authentic user-supplied screenshot; original interface and filter state retained.

Figure 12 shows 10,000 customers, 20.37% churn, 79.63% retention and 4,849 inactive customers. France, Germany and Spain and all four product counts are selected. The activity filter is All. These values reconcile with the chart exports and establish the full-population dashboard denominator.

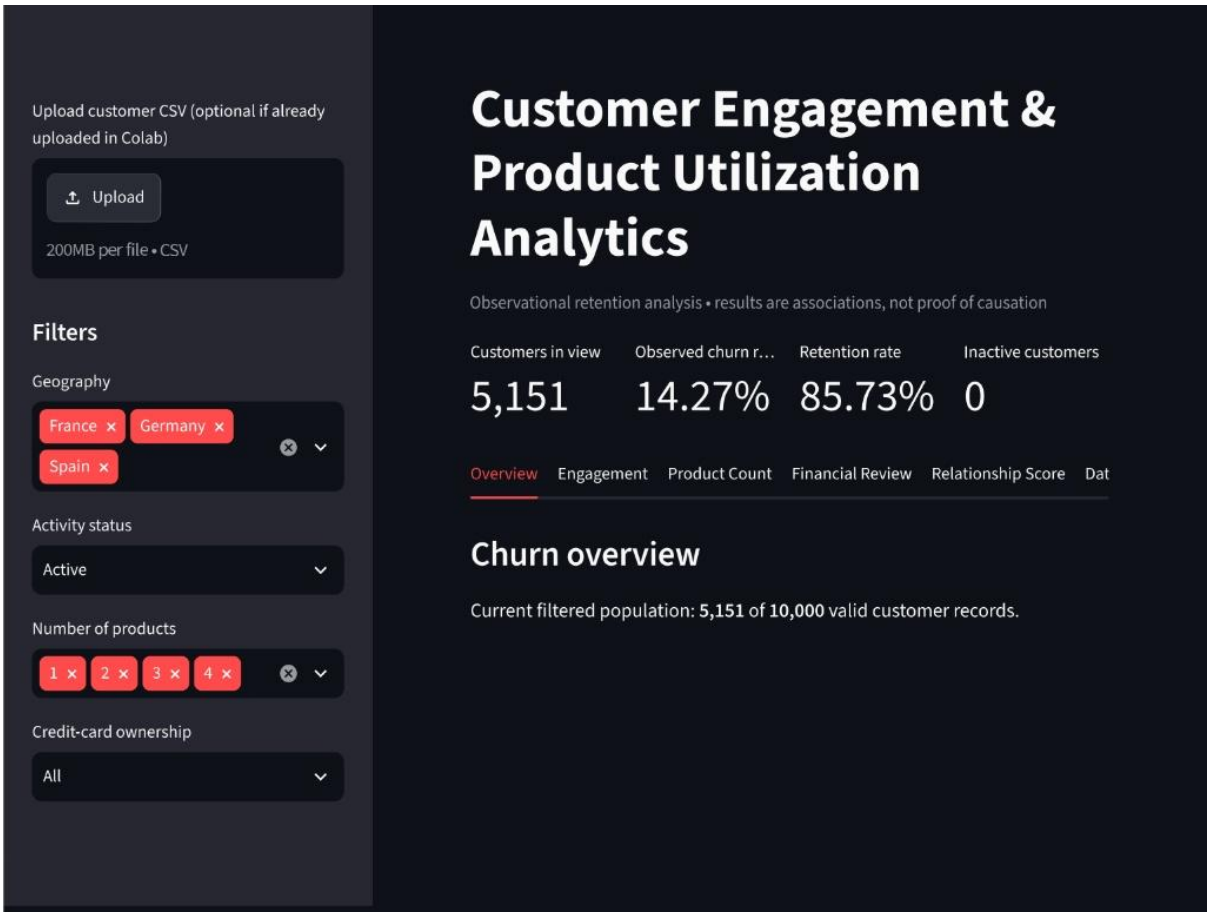


Figure 13: Supplied dashboard overview: active customers

Source: authentic user-supplied screenshot; original interface and filter state retained.

Figure 13 shows 5,151 customers, 14.27% churn, 85.73% retention and zero inactive customers after the Active filter is applied. The inactive KPI is therefore a count within the filtered view, not a fixed whole-population value. This distinction is essential when users compare screenshots.

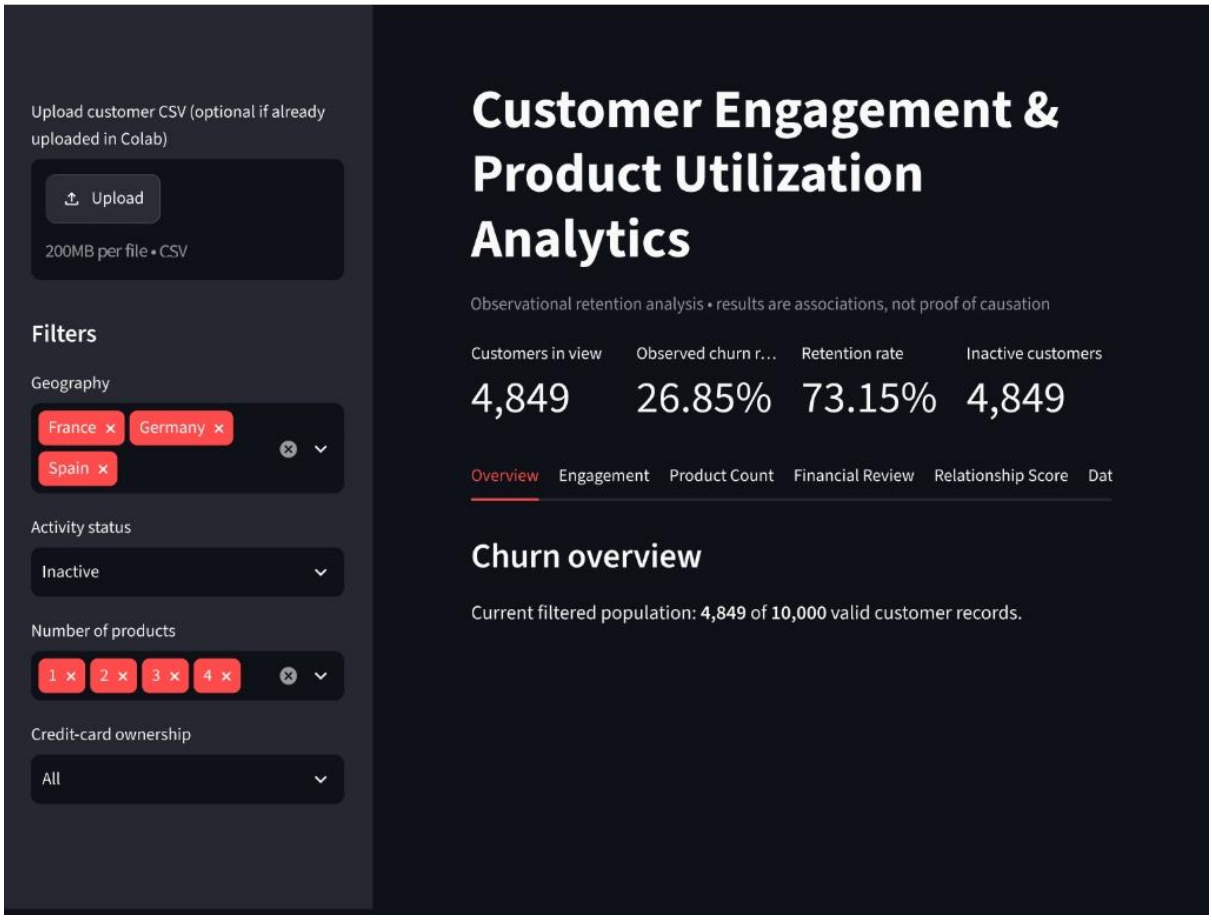


Figure 14: Supplied dashboard overview: inactive customers

Source: authentic user-supplied screenshot; original interface and filter state retained.

Figure 14 shows 4,849 customers, 26.85% churn and 73.15% retention, with all 4,849 customers classified as inactive in the selected view. The three overview states are complementary and numerically consistent; they are not three different reporting periods.

17.2 Chart interpretation and implementation limits

Appendix C retains the three accompanying chart excerpts as supplied. They show activity and product-count panels and explicit cautions about interpreting high churn. In the all-customer excerpt, only the inactive activity bar and the first two product bars are visible. The active and small high-product groups are not established as absent from the analysis: they may be outside the captured viewport or clipped in the interface. The complete HTML outputs include both activity categories and all four product categories.

Counts printed inside the screenshot product bars are segment denominators, while the vertical axis measures churn rate. They must not be read as churn counts. Without a running application, it is not possible to diagnose the clipping, inspect tab contents, verify hover behavior or confirm download features. No screenshots were synthesized and no claim is made that six distinct requested dashboard pages were captured. Operational exports should retain filters and show all category labels.

Chapter 18: Business Recommendations

The recommendations are evidence-linked hypotheses for investigation and evaluation. Priority refers to review sequencing, not a validated ranking of expected financial return. Each proposed intervention needs future outcome data and an appropriate comparison group; no specific churn reduction is promised.

Table 18: Evidence-linked retention recommendations and measurement plan

Priority	Target segment	Supporting evidence	Proposed action	Expected mechanism	Monitoring KPI	Measurement approach	Limitations
1	Three / four products	82.71% / 100% churn; n = 266 / 60	Audit coding, product mix and exit timing	Identify source defects or product-fit concerns	Verified outcomes; complaints; product activity	Source audit, then prospective review with comparison	Small groups; rule-group n differs; no causal result
1	Inactive, high balance	985 customers; 30.05% churn	Consent-aware relationship and service review	Discover unmet needs; relevance is a hypothesis	Subsequent churn; reactivation; contact burden	Randomized outreach versus usual service, fixed horizon	Balance is not profit; contact can cause burden
2	Inactive customers	26.85% vs 14.27% active churn	Diagnose inactivity and test service support	Resolve barriers if those barriers are actionable	Transactions; active use; later retention	Randomized pilot; intent-to-treat; harm monitoring	Binary activity has unknown timing and definition
2	One-product customers	27.71% vs 7.58% with two products	Needs-led product-fit conversation	Offer relevant services, not product accumulation	Actual usage; satisfaction; churn	Randomize eligible offers; compare net outcomes	Selection confounding; more products not uniformly better
3	Cardholders / non-holders	Retention gap +0.63 points; p = .475	Collect usage data before card-led targeting	Test whether use adds information beyond ownership	Frequency; recency; complaints; retention	Validate incremental signal, then controlled test	Ownership is not spending or engagement
Enabler	All analytical segments	Limited behavioral and temporal data	Add usage, costs, outcome dates and governance	Improve measurement, validity and evaluation	Coverage; data quality; fairness; incremental outcomes	Versioned pipeline; temporal validation; controlled pilots	Access, privacy, data minimization and drift

Before deployment, predefine eligibility, consent, randomized assignment, outcome horizon and harm monitoring. Assess subsequent retention, complaints and incremental costs. High observed risk is not equivalent to intervention responsiveness (Ascarza, 2018). See Chapter 20 for evaluation requirements.

Chapter 19: Limitations

19.1 Data provenance and measurement

The most important limitation is incomplete primary evidence. The raw CSV, notebook, output ZIP and dashboard source were not attached. Internal chart consistency cannot prove the absence of raw missing values, duplicate customers, invalid categories, outcome miscoding or synthetic data artifacts. The expected 14-column schema and possible zero-missingness assertions in the brief are not independently validated. Reporting period, event window, Year constancy and the definition of active membership remain unknown.

Recorded activity is a binary proxy, product count does not reveal utilization, and card ownership does not reveal card use. Balance is not profitability or lifetime value. Estimated salary is not verified income or a direct measure of assets. The salary chart includes only 3,000 observations and its sampling method is unavailable. The reporting of financial units deliberately avoids an unsupported currency assumption.

19.2 Design, uncertainty and inference

The observational design does not establish that inactivity, product holdings, geography or age causes exit. Confounding, reverse timing and selection are unresolved. There are no explicit churn dates, exposure intervals, longitudinal histories or experimental outcomes. Thus, the paper cannot estimate survival, annual transitions, campaign efficacy or policy effects.

Small three- and four-product groups yield greater uncertainty and warrant provenance checks. Multiple exploratory comparisons can generate chance findings; Holm adjustment addresses the nine declared omnibus tests only. Confidence intervals rely on statistical assumptions, do not account for unknown customer clustering and cannot correct a non-representative sample. Statistical significance is not commercial importance. Univariate association strength is not predictive feature importance.

19.3 Scores, dashboard and external validity

The RSI is a heuristic with unavailable implementation details and no out-of-sample validation. Monotonic score-band rates do not establish individual calibration, fairness or stability. Segment-label precedence is unclear; the 288-versus-326 high-product discrepancy demonstrates why rules must be inspected. The screenshot set documents three filter states, not six separate dashboard pages, and the application was not launched.

Results cannot be generalized to all European banks, national markets or public-sector populations. No guaranteed retention gain, savings estimate, validated churn probability or automatic customer-treatment rule is supported. What is supported is a set of internally reconciled output-level associations and concrete questions for validation and evaluation.

Chapter 20: Future Research and Additional Data Requirements

20.1 Build an event-time customer evidence base

A stronger analytical dataset should link permissioned monthly activity histories to explicit product and churn events. Include digital-login frequency, transaction recency and frequency, deposits and withdrawals, product acquisition and closure dates, active product usage, card transactions, service contacts, complaints and customer satisfaction. Add validated revenue, fees and cost-to-serve when an economic decision is intended. Keep estimated and verified financial attributes distinguishable.

Outcome timestamps and a stable definition of exit are prerequisites. Features must be measured before the prediction date and outcomes after it to avoid leakage. A single constant reporting year, if later confirmed, would provide no within-dataset year-over-year comparison; future collection should preserve multiple periods. Missingness, changes in product coding and account-to-customer linkage require ongoing audit rather than a one-time check.

20.2 Prediction and score validation

With appropriate longitudinal data, compare transparent baselines with well-specified predictive models using customer-level train / validation separation and time-based holdouts. Evaluate discrimination, precision–recall behavior, calibration and decision costs, not accuracy alone (Neslin et al., 2006). Check performance and service consequences across relevant groups and reporting periods. Test whether the RSI adds useful information beyond activity and product count separately; examine weight sensitivity and high-product exceptions.

Model selection should be guided by the intended decision, with realistic costs and outcome values when available (Verbeke et al., 2012). No algorithm is specified as superior in advance. Changes in population or service offerings require drift monitoring and possible recalibration.

20.3 Establish intervention effectiveness separately

Prediction identifies who is likely to leave under observed conditions; causal evaluation asks what changes the outcome. Collect marketing assignment, actual contact, delivery, opt-outs and later outcomes in controlled trials. Predefine the primary endpoint, follow-up horizon and eligible population. Report intent-to-treat differences with uncertainty, implementation failures and customer harms. An uplift or treatment-effect model should be considered only after credible treatment and outcome evidence exists.

Future public-sector work would additionally need a documented sampling frame, appropriate market coverage and justified aggregation before making population-level claims. Research partnerships should use data minimization, role-based access and carefully controlled release of aggregate findings.

Chapter 21: Conclusion

The supplied outputs consistently describe 10,000 customer observations, 2,037 churns and 7,963 retained outcomes. Recorded inactivity is associated with higher churn: 26.85% compared with 14.27% among active customers. The activity gap is large enough to merit investigation, but it does not establish that outreach or reactivation will prevent exit.

Product breadth has a non-linear relationship with observed churn. Two-product customers have the lowest churn at 7.58%, versus 27.71% with one product, while the three- and four-product categories show 82.71% and 100.00%. Small segment sizes, unknown product combinations and incomplete source validation require caution. Increasing product count cannot be recommended as a universal retention mechanism.

Financial commitment and engagement provide complementary information. The reconstructed high-balance group contains 985 inactive observations, with 30.05% churn; substantial recorded funds do not establish loyalty or profitability. Card ownership shows only a small, statistically inconclusive retention difference. Geography and age bands vary markedly, whereas tenure and credit-score band comparisons are inconclusive in the declared test family. Those supporting patterns do not identify causal or discriminatory treatment criteria.

Transparent review groups and the four RSI bands differentiate observed outcomes, and score-band churn decreases monotonically. Nevertheless, the score is not a validated probability and the segment hierarchy requires source inspection. Dashboard screenshots demonstrate consistent all / active / inactive denominators and the value of visible filter context, while also revealing the limits of clipped chart excerpts.

The ten research questions are therefore answered descriptively where evidence exists: activity, individual product categories, non-linearity, high-balance inactivity, card ownership, supporting characteristics, reported segments and score bands all have quantified output-level findings. Exact joint assignment rules and intervention effectiveness remain open. The defensible next steps are a complete source audit, richer behavioral and temporal measurement, and controlled service-oriented trials. Engagement and product-relationship analytics can support better-informed retention reviews, but operational deployment requires privacy safeguards, validation and evidence of customer benefit.

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Bibliographic existence and metadata verified against Crossref DOI records and publisher / institutional search records on 23 October 2026. Literature is used for conceptual and methodological claims, not as evidence of this dataset's outcomes.

Primary project evidence

User-supplied, unpublished project materials (undated): the master research brief; twelve Plotly HTML analytical exports, cited by original filename in figure source notes; and six dashboard screenshots. The original customer dataset's provenance, author and observation period are not documented in the supplied attachments. The paper does not substitute a similar public churn dataset.

Appendix A: Additional analytical graphs

All 12 supplied analytical graphs are included once: ten appear in the main results chapters and the following two supporting figures complete the set. Formatting has been standardized while retaining the embedded observations, categorical membership and numerical results. These figures supplement Chapter 13 and are not additional independent samples.

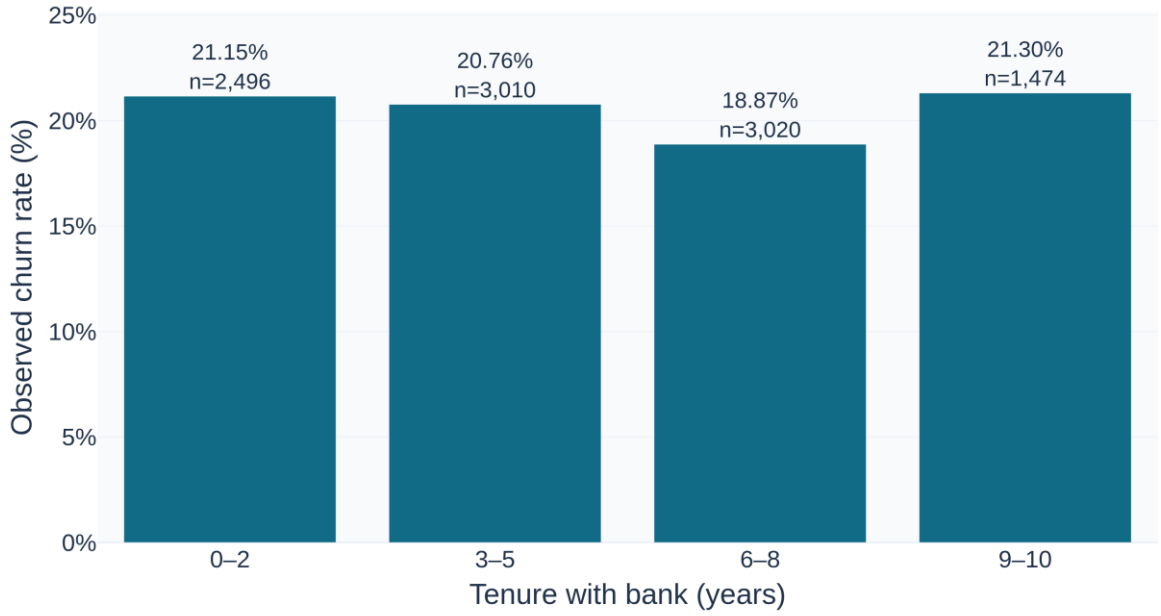


Figure 15: Observed churn by tenure band

Source: supplied churn_by_tenure_band.html; embedded Plotly data rendered at 3,000 × 1,700 pixels. Visual formatting standardized; observations unchanged.

Figure 15 shows modest differences across tenure bands: 21.15%, 20.76%, 18.87% and 21.30%. The lowest observed rate is in the six-to-eight-year band, but the omnibus association is inconclusive after multiplicity adjustment. Neither a linear loyalty effect nor equivalence across bands is established.

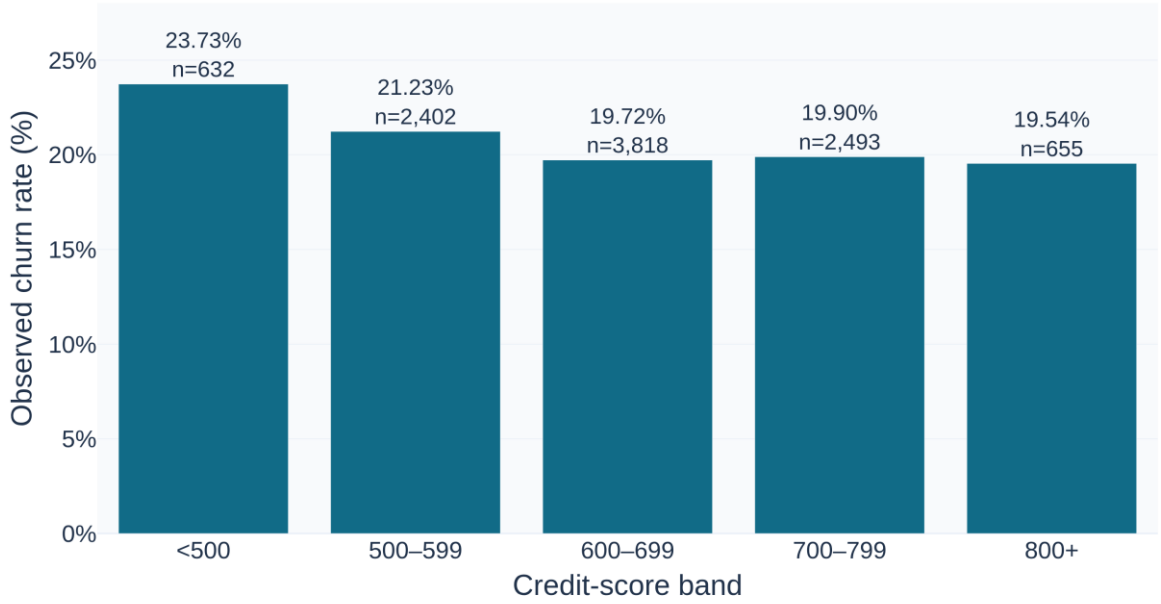


Figure 16: Observed churn by credit-score band

Source: supplied churn_by_credit_score_band.html; embedded Plotly data rendered at 3,000 × 1,700 pixels. Visual formatting standardized; observations unchanged.

Figure 16 shows the highest churn in the below-500 band (23.73%, n = 632), declining to roughly 20% in the upper bands. Overlapping uncertainty and the inconclusive omnibus test limit interpretation. Credit-score band is not treated as a service eligibility rule or a validated retention predictor.

Appendix B: Supplementary results tables

B.1 Principal contrasts and denominator checks

Table 19: Supplementary contrasts and denominator definitions

Measure	Estimate	Interval / denominator
Overall churn	20.37%	2,037 / 10,000
Inactive minus active churn	+12.58 percentage points	95% approximate CI 11.01–14.15 points
Inactive-to-active rate ratio	1.88	95% log interval 1.73–2.04
One minus two product churn	+20.13 percentage points	1,409 / 5,084 versus 348 / 4,590
Cardholder retention difference	+0.63 percentage points	95% approximate CI −1.11 to +2.37 points
High-balance disengagement	49.25%	985 inactive / 2,000 high balance
Inactive high-balance population share	9.85%	985 / 10,000
Mean product count	1.5302	15,302 total product holdings / 10,000

B.2 Supporting-variable outcome counts and intervals

Table 20: Supporting partitions: recovered outcomes and Wilson uncertainty

Partition / group	Exited	Retained	95% CI: churn
Geography / France	810	4,204	15.16–17.20%
Geography / Germany	814	1,695	30.64–34.30%
Geography / Spain	413	2,064	15.26–18.19%

Partition / group	Exited	Retained	95% CI: churn
Age band / 18–29	124	1,517	6.37–8.94%
Age band / 30–39	473	3,873	9.99–11.84%
Age band / 40–49	806	1,812	29.05–32.58%
Age band / 50–59	487	382	52.72–59.31%
Age band / 60+	147	379	24.28–31.93%
Tenure band / 0–2	528	1,968	19.60–22.80%
Tenure band / 3–5	625	2,385	19.35–22.25%
Tenure band / 6–8	570	2,450	17.52–20.31%
Tenure band / 9–10	314	1,160	19.29–23.47%
Credit-score band / <500	150	482	20.58–27.20%
Credit-score band / 500–599	510	1,892	19.64–22.91%
Credit-score band / 600–699	753	3,065	18.49–21.01%
Credit-score band / 700–799	496	1,997	18.38–21.51%
Credit-score band / 800+	128	527	16.69–22.75%

Counts were recovered from full-precision embedded rates and source segment sizes, then checked against each partition's totals. Intervals were newly computed for the paper. Full product, activity and score-band outcome counts appear in their main-chapter tables. The raw-product categories and the labeled high-product segment are deliberately not conflated.

Appendix C: Streamlit dashboard screenshots

The following excerpts preserve the remaining three supplied images. Together with Figures 12–14, every supplied dashboard screenshot is embedded in the paper. They document the chart panels corresponding to the all / active / inactive filter states, not distinct financial, score or data-quality pages. The empty or clipped areas are retained as source evidence rather than reconstructed.

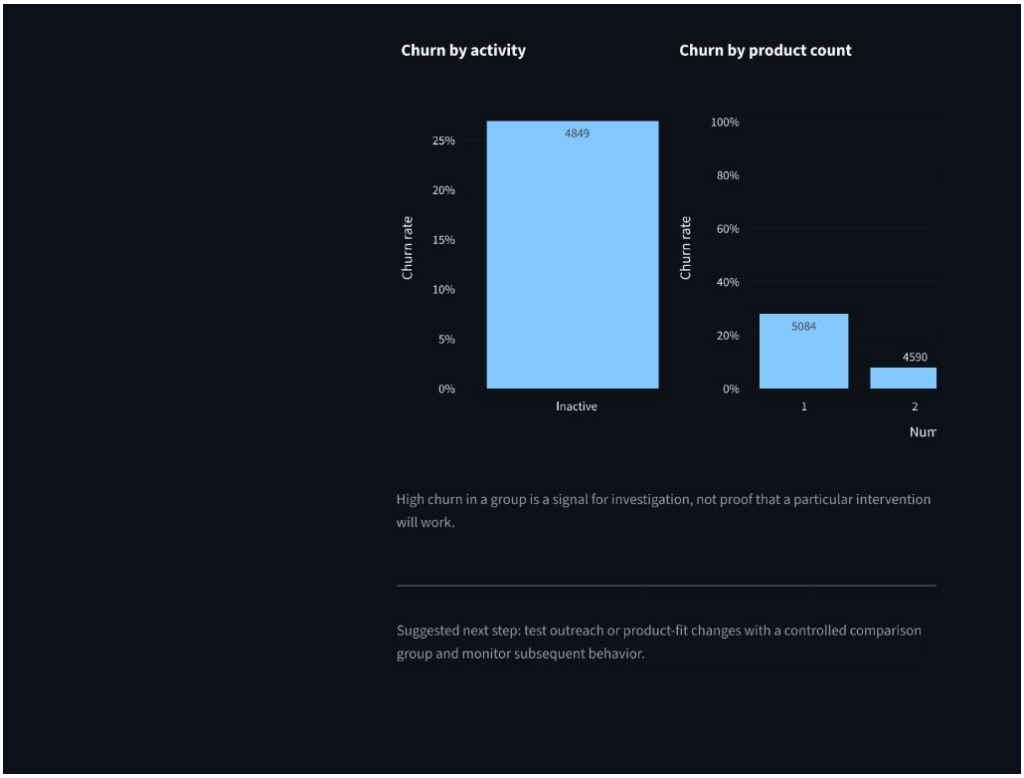


Figure 17: Supplied dashboard chart excerpt: all-customer state

Source: authentic user-supplied screenshot; original interface and filter state retained.

Figure 17 shows the inactive bar near 26.85% and the first two product bars, labeled with customer counts 5,084 and 4,590. Only part of the categorical display is visible. The full HTML product chart, rather than this excerpt, establishes the three- and four-product results. The interface text appropriately recommends controlled comparisons rather than causal interpretation.

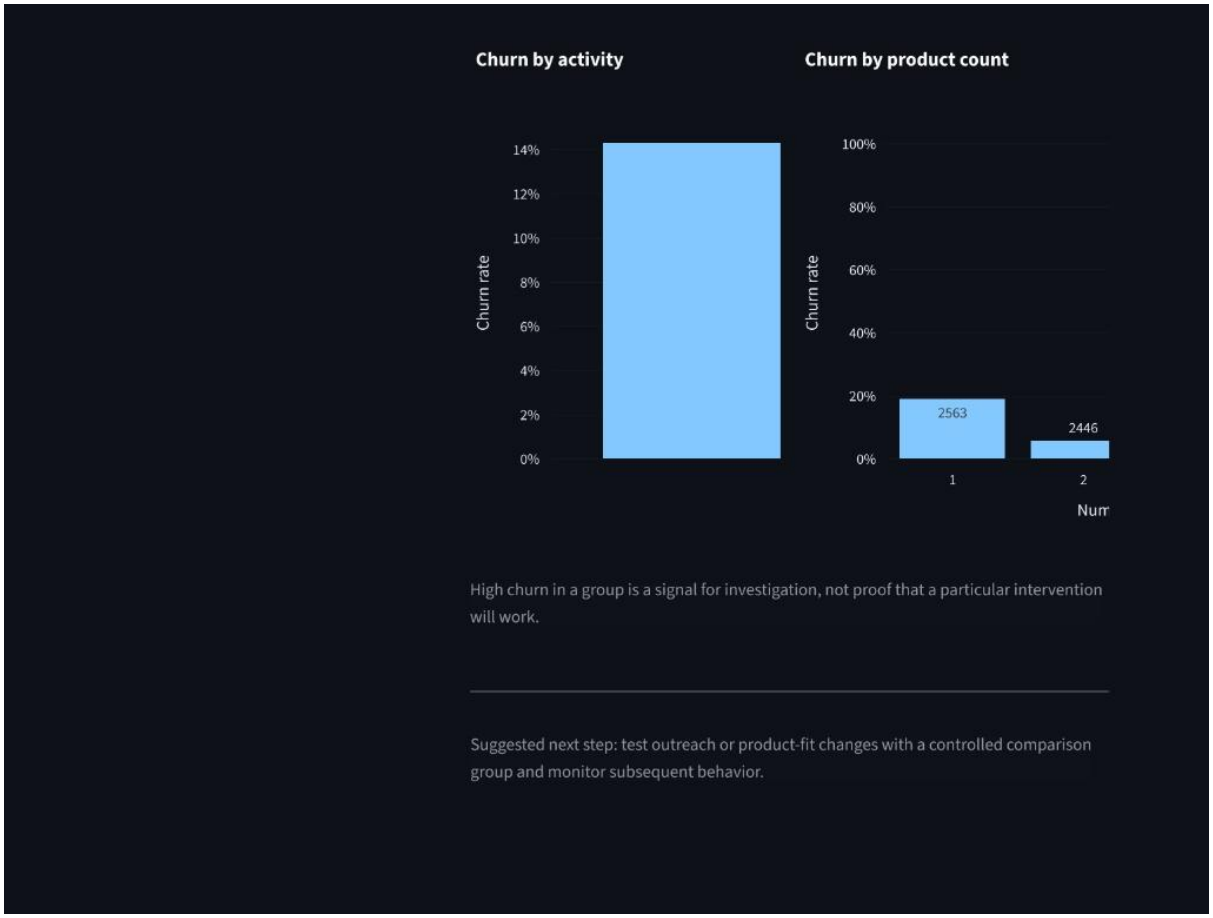


Figure 18: Supplied dashboard chart excerpt: active-customer state

Source: authentic user-supplied screenshot; original interface and filter state retained.

Figure 18 accompanies the active overview: the activity bar is near 14.27%, and the visible one- and two-product denominators are 2,563 and 2,446. These are filtered counts, not full-population values. The truncated labels and viewport limit numerical interpretation beyond the displayed elements.

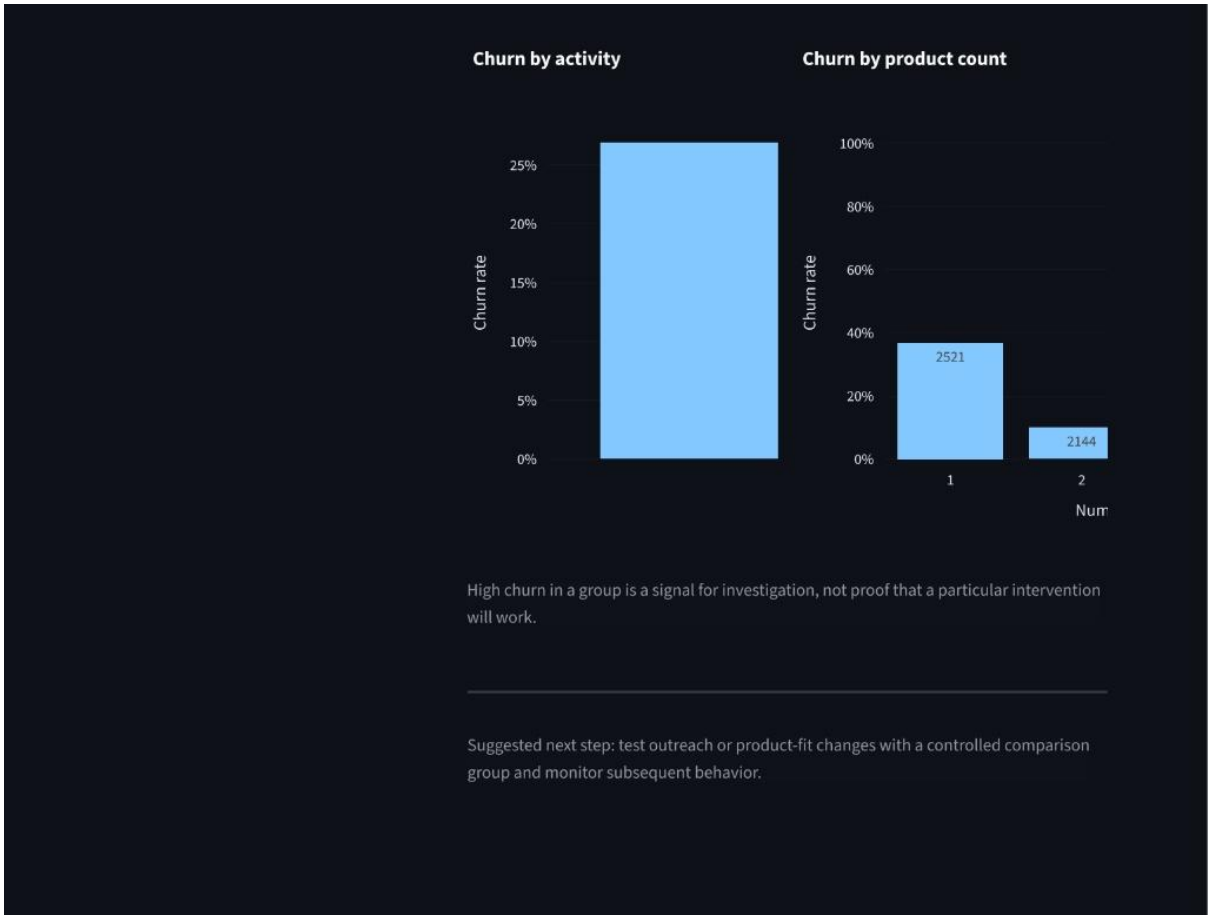


Figure 19: Supplied dashboard chart excerpt: inactive-customer state

Source: authentic user-supplied screenshot; original interface and filter state retained.

Figure 19 accompanies the inactive overview: the activity bar is near 26.85%, with visible one- and two-product denominators of 2,521 and 2,144. The one-product counts across active and inactive views sum to 5,084, and the two-product counts sum to 4,590, providing a useful screenshot-level consistency check. Missing on-screen categories are not assumed to have zero customers.

Appendix D: Methodology and reproducibility

D.1 Reproducing the available-evidence analysis

1. Preserve all 12 original HTML files and six screenshots; record filenames and evidence provenance. Treat self-contained chart exports as potentially sensitive because they contain observations, not just pixels.
2. Decode the final Plotly.newPlot call in each HTML file and preserve data and layout arrays. Recover segment counts, full-precision rates and custom-data fields without parsing the bundled JavaScript library as analytical content.
3. For each rate category, multiply denominator by rate, recover the integer event count and check exact-rate consistency. Confirm every categorical partition totals 10,000 and 2,037 churns.
4. Use outcome-specific product histograms to cross-check all four product counts. Extract paired activity and balance values within each outcome trace, without linking separate chart arrays by position.
5. Recompute financial summaries and the population balance 80th percentile with linear interpolation. Apply the unrounded threshold inclusively and compare the resulting 985 inactive high-balance observations and 296 churns with the source priority chart.
6. Calculate Wilson 95% intervals, uncorrected Pearson tests and Cramér's V. Adjust the nine omnibus p-values with Holm. Label supplementary contrasts and the Fisher test separately.
7. Render the actual Plotly data as static images; retain source values and use readable axes, counts, legends and source notes. Use original screenshots without synthesizing missing pages.
8. Regenerate the Word paper, update its table of contents, reopen it and inspect rendered pages for clipping, blank pages, table overflow and broken figures.

D.2 Evidence-to-calculation manifest

Table 21: Reproduction manifest for the supplied output files

Original file(s)	Evidence recovered	Use in paper
churn_by_activity.html	Activity n, rates, intervals	Engagement contrasts and Figure 2
churn_by_product_count.html; customer_count_by_product_and_churn.html	Product rates and raw outcome-count arrays	Four-category audit and Figures 3–4
balance_activity_churn.html	10,000 activity / balance / outcome observations	Financial summaries, threshold and Figure 5
salary_vs_balance.html	3,000 displayed points with activity and outcome	Sample-only interpretation and Figure 6
churn_by_card_ownership.html	Card ownership denominators and rates	Ownership contrast and Figure 7
churn_by_geography.html; churn_by_age_band.html	Supporting group sizes and rates	Figures 8–9 and association tests
retention_priority_by_churn.html	Five labels, counts, rates and mean balances	Figure 10 and assignment discrepancy

Original file(s)	Evidence recovered	Use in paper
relationship_strength_by_churn.html	Four band counts, rates and mean balances	Figure 11; no individual scores
churn_by_tenure_band.html; churn_by_credit_score_band.html	Supporting band counts and rates	Figures 15–16; inconclusive tests

D.3 Outstanding verification and missing requested assets

Table 22: Items not supplied or not verifiable in this edition

Requested evidence / analysis	Status and implication
Raw European_Bank.csv or equivalent	Missing; schema, duplicates, missingness, raw ranges and original exclusions cannot be audited
Colab notebook and generated-output ZIP	Missing; original code, library stack and workflow not inspected
Streamlit app source and running environment	Missing; exact segment precedence, score normalization, threshold configuration and downloads unverified
Six distinct dashboard-page screenshots	Not supplied; six available images show three overview states and three chart excerpts only
Year and reporting period	Unavailable; no year-over-year or annualized interpretation
Gender analysis	Unsupported; omitted rather than inferred
Full segment medians / product mix / activity composition	Unavailable for reported rule labels; no invented supplementary values
Full-population salary joint analysis	Only 3,000 displayed points available; no population salary-based segment counts
Predictive / causal validation and business outcomes	No training, holdout evaluation, intervention effects, profit or savings claims

To reproduce the original project rather than this output-based study, obtain the customer CSV, notebook, aggregate tables and application repository, document their versions, and reconcile them against the exported counts before changing the findings. Investigate every discrepancy openly. The present edition includes all available graphs and screenshots, while its missing-evidence register preserves the distinction between an observed output and an independently validated analytical pipeline.